Classical recording: complete memories, refinement and nonlinear limits

A supplied positive Gaussian covariance restriction is preserved by affine canonical dynamics and complete terminal coordinate memories. A bounded physical feedback policy has a partition-independent continuous body–record limit, with controller recoil retained and the conditional floor surviving its full declared coordinate record. The same construction admits zero action. Moreover, granting passive earlier readouts or nonlinear body copies defeats the proposed unrestricted classical floor. A fixed-cutoff common radiation bath does not repair that observation class.

These results concern specified classical models and records. They separate existence of dynamics, conditional action floors, a supplied quantum reference and identification of a universal scale with physical $\hbar$. The nonlinear quantum-reference curvature and phase tests do not derive that reference. All model/coordinator checks recorded during development were internal; corrections are incorporated in the statements below.

Every covariance-restricted theorem below assumes the **joint** Gaussian condition, with the canonical Poisson matrix Ω_n,

$$V + i\kappa\Omega_n \succeq 0. \tag{22}$$

Marginal one-pair floors alone do not imply this condition. Its calibration $\kappa = \hbar/2$ is an added identification; the radiation-unit and SED qualifications are in unit and indeterminacy and the resonant link.

The score-constrained treatment contains the shared-sign preparation, its exact copy law, terminal-conjugate escape, free transport failure and switching realization. Those proofs are maintained there once. Here κ is a supplied action scale; a cutoff, record precision, feedback gain, time mesh and $\epsilon = \hbar/S_0$ are distinct parameters.

The regulated common bath and the instrument class

Use SI units, $\kappa > 0$, speed c, and three harmonically bound dipole coordinates q_i, p_i, $i = 0, 1, 2$, with masses $m_i > 0$, frequencies $\omega_i > 0$, charges $e_i \neq 0$, fixed centres $\mathbf{r}_i$, and unit directions $\mathbf{n}_i$. The body has index 0. The traps and external quadratic controls define the linear approximation. The incident Gaussian field has the zero-point spectrum

$$u_0(\omega) = \frac{\kappa\omega^3}{\pi^2 c^3}, \qquad \mathcal{E}_0(\omega) = \kappa\omega. \tag{1}$$

Introduce a real common form factor $f_\Lambda(\omega)$, equal to one on the frequencies of interest and zero for $\omega > \Lambda$, with $\Lambda < \infty$. This can regulate the incident force itself or the dipole coupling to the field. In the latter convention the full incident field retains (1), while the apparatus response selects a frame and a frequency band. All covariances below refer to the regulated response.

The angular correlation matrix is

$$K_{ij}(\omega) = \frac{3}{8\pi}\int_{S^2}\Big[\mathbf{n}_i\cdot\mathbf{n}_j - (\mathbf{n}_i\cdot\widehat{\mathbf{k}})(\mathbf{n}_j\cdot\widehat{\mathbf{k}})\Big]\cos\!\left(\frac{\omega}{c}\widehat{\mathbf{k}}\cdot(\mathbf{r}_i - \mathbf{r}_j)\right) d\widehat{\mathbf{k}}. \tag{2}$$

It is positive semidefinite and $K_{ii} = 1$: the transverse polarization sum is a Gram matrix, and the integral for a unit direction is $8\pi/3$. Thus the same field gives both diagonal and cross force covariances. Define its matrix spectral density by

$$J_{ij}(\omega) = \frac{e_i e_j \omega^3}{6\pi\epsilon_0 c^3} f_\Lambda(\omega)^2 K_{ij}(\omega),$$

$$C_F(t-s) = \mathbb{E}[F(t)F(s)^{\mathsf{T}}] = \frac{2\kappa}{\pi}\int_0^\Lambda J(\omega)\cos[\omega(t-s)]\,d\omega. \tag{3}$$

For example $C_{F,ii}(0) = e_i^2\kappa\int_0^\Lambda \omega^3 f_\Lambda^2 d\omega/(3\pi^2\epsilon_0 c^3)$. The normalization follows directly from $u_0 = \epsilon_0\langle|\mathbf{E}|^2\rangle$ per frequency, isotropy, and $F_i = e_i E_i$.

Keep radiation damping through a passive oscillator bath with the same J, including the cross damping. Its causal friction kernel is

$$\mu(t) = \frac{2}{\pi}\int_0^\Lambda \frac{J(\omega)}{\omega}\cos(\omega t)\,d\omega, \qquad t \geq 0. \tag{4}$$

An explicit realization uses harmonic bath coordinates coupled linearly to the q_i, with the quadratic counterterm obtained by completing each bath potential square. Choose the bath coupling matrix so that $J(\omega) = \pi L(\omega)L(\omega)^{\mathsf{T}}/(2\omega)$; its existence follows from positivity of (2). Give a free bath coordinate of frequency ω the variances κ/ω and $\kappa\omega$ for coordinate and momentum. Its energy is $\kappa\omega$. Eliminating it gives (3) and (4). The counterterm keeps the prescribed trap stiffness positive; the finite-cutoff bath Hamiltonian has positive quadratic energy.

In this realization $\dot q = \partial H_{\mathrm{sys}}(t)/\partial p$, and

$$\dot p = -\frac{\partial H_{\mathrm{sys}}(t)}{\partial q} - \int_0^t \mu(t-s)\dot q(s)\,ds + F(t). \tag{5}$$

An initial-slip term, or the stationary prehistory, is included in $F(t)$ with its correlations with the initial system. Independent system–bath initial data are unnecessary below. Before the controls, the diagonal absorptive part is the usual radiation term $m_i\tau_i\omega^3 f_\Lambda^2$, where

$$\tau_i = \frac{e_i^2}{6\pi\epsilon_0 m_i c^3}. \tag{6}$$

This finite-cutoff memory equation retains the dispersive part as well. It avoids using a third-order local radiation equation during arbitrarily fast controls. Static dipole interactions, when retained, are additional finite entries of the stiffness matrix; assume stability of the chosen trapped system.

Tracking the prior. Set

$$Q_i = \sqrt{m_i\omega_i}\,q_i, \qquad P_i = \frac{p_i}{\sqrt{m_i\omega_i}}, \qquad Z = (Q, P, Y, \Pi, Z_2, \Pi_2)^{\mathsf{T}}. \tag{7}$$

Here $(Q,P) = (Q_0,P_0)$, $(Y,\Pi) = (Q_1,P_1)$ and $(Z_2,\Pi_2) = (Q_2,P_2)$. Every coordinate has units $\sqrt{\text{action}}$, every pair remains canonical, and the body's covariance determinant equals that in (q_0,p_0). For a stable stationary response, let $H(\omega)$ be the six-by-three transfer matrix from physical forces to these variables. The complete prior covariance is

$$V = \mathrm{Re}\int_0^\Lambda H(\omega)\frac{2\kappa J(\omega)}{\pi}H(\omega)^\dagger\,d\omega. \tag{8}$$

The susceptibility in H includes (4); momentum rows before the controls are $-im_i\omega$ times the coordinate rows before scaling. Any undamped initially populated normal modes add their own covariance. Formula (8) displays the cross correlations explicitly, and the theorems use the full V of (8), cross correlations included. Its dispersive cross part and the constant K of (14) are cutoff constants (σ_P^2 grows with Λ). The radiation time here is the SI $\tau = e^2/(6\pi\epsilon_0 mc^3)$; the SED link note uses the Gaussian-unit form.

For an isolated weakly damped resonance the ideal narrow-resonance limit gives $\mathrm{Var}\,Q_i = \mathrm{Var}\,P_i = \kappa$. A finite spectral window and finite damping introduce corrections; the exact finite-cutoff statements below allow all of them. In particular, a literal window of only a fixed number of linewidths captures only a fraction of the resonant Lorentzian. Access to a temporally filtered variable also requires a recording prescription; the instantaneous pulse proof is applied to canonical cutoff variables, or to canonical oscillator amplitudes in the stated resonant model.

Passive retained readings defeat a prior-only restriction

Assume Z has a centered Gaussian law with finite covariance V; known means can be subtracted. Grant two quadratic Hamiltonian pulses, their integrated Hamiltonians being

$$\mathcal{H}_1 = GQ\Pi, \qquad \mathcal{H}_2 = B\Pi\Pi_2, \qquad G, B > 0. \tag{9}$$

Both have action units. Their Hamilton equations are linear. The first pulse sends $Y \mapsto Y + GQ$ and $P \mapsto P - G\Pi$. Read and store $R_1 = Y + GQ$ before the second pulse. That pulse sends $Z_2 \mapsto Z_2 + B\Pi$ and $Y \mapsto Y + B\Pi_2$; it leaves the body and Π fixed. Read $R_2 = Z_2 + B\Pi$. The final body is

$$X = (Q, P - G\Pi)^{\mathsf{T}}, \qquad R = (R_1, R_2)^{\mathsf{T}}. \tag{10}$$

The readouts are classical passive coordinate observations. The first stored number survives the later change of Y. Small independent Gaussian readout errors can also be allowed, as in the finite-duration construction.

Theorem 1. Under these instrument assumptions, for every such V,

$$\boxed{\sqrt{\det \operatorname{Cov}(X \mid R)} \le \frac{\sigma_Y \sigma_P}{G} + \frac{\sigma_Y \sigma_{Z_2}}{B}.} \tag{11}$$

Here $\sigma_U^2 = \operatorname{Var} U$ in the initial joint state. In particular the posterior area has infimum zero as $G, B \to \infty$. All cross correlations in V are allowed, including perfect ones.

Proof. Estimate the final body by

$$\widehat{X} = (R_1/G, -GR_2/B)^{\mathsf{T}}.$$

The errors are the exact random-variable identities

$$X - \widehat{X} = (-Y/G,\ P + GZ_2/B)^{\mathsf{T}}. \tag{12}$$

Thus the first pointer's recoil Π cancels, including the part correlated with the body or either pointer. The conditional Gaussian covariance is independent of the outcome. Conditional expectation minimizes the error covariance in the positive-semidefinite order, so it is bounded above by the covariance of (12). In particular,

$$\operatorname{Var}(Q \mid R) \le \sigma_Y^2/G^2, \qquad \operatorname{Var}(P - G\Pi \mid R) \le (\sigma_P + G\sigma_{Z_2}/B)^2.$$

The second bound is Cauchy–Schwarz, with either sign of $\operatorname{Cov}(P, Z_2)$. Bounding a two-by-two determinant by the product of its diagonal entries proves (11). □

For direct use of a specified common-field response, introduce

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -G & 0 & 0 \end{pmatrix}, \qquad L = \begin{pmatrix} G & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & B & 1 & 0 \end{pmatrix}.$$

The exact answer, with every entry from (8) retained, is

$$\Sigma_{\text{post}} = AVA^{\mathsf{T}} - AVL^{\mathsf{T}}(LVL^{\mathsf{T}})^{-1}LVA^{\mathsf{T}}. \tag{13}$$

Use a pseudoinverse on the supported record subspace when necessary. For example the two record covariances include

$$\operatorname{Cov}(R_1, R_2) = GV_{QZ_2} + GBV_{Q\Pi} + V_{YZ_2} + BV_{Y\Pi}.$$

The covariance of the error vector in (12) has off-diagonal entry $-V_{YP}/G - V_{YZ_2}/B$ and second diagonal entry $V_{PP} + 2GV_{PZ_2}/B + G^2V_{Z_2Z_2}/B^2$. These formulas keep every cross entry of V. Equation (11) bounds their most unfavourable possible effect.

Explicit constants. If $\sigma_Y^2, \sigma_P^2, \sigma_{Z_2}^2 \le K\kappa$, put $B = G^2$. Then

$$\sqrt{\det \Sigma_{\mathrm{post}}} \le K\kappa(G^{-1} + G^{-2}). \tag{14}$$

Every finite-cutoff prior has a finite such K. Taking $G > 2K + 1$ gives a strict bound below κ. For ideal resonant marginals, $K = 1$, and $G = 2$, $B = 4$ give $3\kappa/4$, regardless of the cross correlations. As an algebraic example only, for $V = \kappa I_6$,

$$\Sigma_{\mathrm{post}} = \begin{pmatrix} \kappa/(1+G^2) & 0 \\ 0 & \kappa[1 + G^2/(1+B^2)] \end{pmatrix}. \tag{15}$$

With $G = 2$, $B = 4$, its area is $\kappa\sqrt{21/85}$. Independence is used only in this example. The body momentum's unconditional spread grows with G; the completed record estimates its kick. The theorem concerns posterior concentration and leaves the unconditional disturbance cost to its own accounting.

Fix Λ, the initial finite Gaussian covariances, and finite gains G, B. Apply the two pulses over successive intervals of total length T, using smooth fixed pulse shapes rescaled by $1/T$. Keep (5) active throughout and retain the first reading at the interval boundary. Let X_T, R_T denote the resulting final body and records. Write their deviations from the ideal expressions (10) as $\Delta Q, \Delta P, \Delta R_1, \Delta R_2$, and set

$$d_Q(T) = \|\Delta Q - \Delta R_1/G\|_2, \qquad d_P(T) = \|\Delta P + G\Delta R_2/B\|_2. \tag{16}$$

Norms are centered root-mean-square norms. They include readout errors when present. The initial field and body can be correlated.

Theorem 2. In the finite-cutoff passive linear model (3)–(5), with the classical readouts of the passive-readout construction, $d_Q(T), d_P(T) \to 0$ as $T \downarrow 0$. Moreover

$$\boxed{\sqrt{\det \mathrm{Cov}(X_T \mid R_T)}} \le \left(\frac{\sigma_Y}{G} + d_Q(T)\right)\left(\sigma_P + \frac{G\sigma_{Z_2}}{B} + d_P(T)\right). \tag{17}$$

Consequently every strict violation in (11) persists for sufficiently small positive T. At $K = 1$, $G = 2$, $B = 4$, the explicit sufficient error budget $d_Q, d_P \le \sqrt{\kappa}/10$ gives area at most $24\kappa/25 < \kappa$.

Proof. At fixed cutoff $C_F(0)$, μ and μ' are finite. In particular

$$\left\| \int_0^T F_i(t)\, dt \right\|_2 \le T\sqrt{C_{F,ii}(0)}.$$

Rapid control-induced changes of q leave the memory term bounded: integration by parts gives

$$\int_0^t \mu(t-s)\dot{q}(s)\, ds = \mu(0)q(t) - \mu(t)q(0) + \int_0^t \mu'(t-s)q(s)\, ds. \tag{18}$$

For the control-only equation each pulse is a finite shear. In the scaled coordinates its propagator and inverse have norms bounded, uniformly in T, by $(1 + G)(1 + B)$. Variation of constants, (18), and the finite force second moment give a uniform mean-square bound on q, p for sufficiently small T by the integral Gronwall inequality. Integrating the remaining trap, bath and memory drifts then makes the endpoint and first-read errors $O(T)$, with constants depending on Λ, G, B, the initial covariances and trap parameters. This proves the asserted convergence, while keeping radiation damping in the equation. The same conclusion holds for any supplied resonant Markov model with finite diffusion, with stochastic increments $O(\sqrt{T})$.

Apply the estimator of the passive-readout construction to R_T. Its two errors equal (12) plus the residuals in (16). The triangle inequality in L^2 gives (17), and Gaussian conditioning again

makes the posterior covariance outcome-independent. Finally $(1/2+1/10)(3/2+1/10) = 24/25$. □

Correlations during the controls. More explicitly, the four-vector $O_T = (X_T, R_T)$ has the form $M_T Z + \eta_T$. If $W_T = \mathrm{Cov}(Z, \eta_T)$ and $N_T = \mathrm{Cov}(\eta_T)$, then

$$\mathrm{Cov}(O_T) = M_T V M_T^{\mathsf{T}} + M_T W_T + W_T^{\mathsf{T}} M_T^{\mathsf{T}} + N_T. \tag{19}$$

For a force-response kernel $D_T(t)$ the driven part is

$$N_T = \int_0^T \int_0^T D_T(t) C_F(t-s) D_T(s)^{\mathsf{T}}\, dt\, ds,$$

with any prehistory terms included in η_T. Thus both the common increments and their correlation with the initial state are retained. Equation (17) uses an L^2 bound, so (17) holds with every cross term of (19) retained. Explicitly, with $\Lambda_* = \sqrt{\pi\omega_0/\tau}$, the budget $d_P \le \sqrt{\kappa}/10$ needs $T\Lambda \le 0.14\,(\Lambda_*/\Lambda)$ from the force noise and $T\Lambda \le 0.15\,(\Lambda_*/\Lambda)^2$ from memory: the pulses must outrun every retained field mode; this estimate is an internal finite-cutoff check and supplies no uniform cutoff-removal construction. The resonant value $K = 1$ used for the $3\kappa/4$ example requires $\Lambda \ll \Lambda_*$; at $\Lambda \sim 1/\tau$, $K \sim 1/(\pi\tau\omega_0)$ and the gains must exceed $2K$.

Small independent Gaussian readout noises of standard deviations r_1, r_2 add at most r_1/G and Gr_2/B to the two budgets (16). Every chosen strict violation therefore permits positive readout noise. A prescribed minimum readout noise, coupling duration, gain ceiling or memory disturbance could restrict the protocol; those would be added instrument premises.

Order and cost of the limits. For a target area, first fix a finite cutoff and its actual prior V, then choose finite gains using (11), then choose a positive duration using (17). The constants depend on the cutoff and the gains. Removing the cutoff before this construction reintroduces the divergent full momentum variance. The impulsive limit also spends unbounded control strength and allows idealized bilinear couplings. This establishes the linear-model obstruction; relativistic locality, material bounds and an implementation using only electromagnetic controls would require further estimates with c. Heavy or detuned pointers are unnecessary for the theorem. Increasing mass can reduce coordinate noise or slow damping, but leaves the equilibrium resonant area κ unchanged.

For a Gaussian posterior, the diagonal bounds (11) also bound centred credible-window half-widths. With $z_\epsilon = \Phi^{-1}(1-\epsilon/2)$,

$$ab \le z_\epsilon^2 \left(\frac{\sigma_Y \sigma_P}{G} + \frac{\sigma_Y \sigma_{Z_2}}{B} \right) \longrightarrow 0. \tag{25}$$

This bounds both marginal windows, even with cross covariance. It can violate the supplied quantum window condition $\lambda_0(ab/\hbar) \ge (1-2\epsilon)^2$ of the fifth-postulate treatment, precisely because these passive retained readings are a different instrument class.

Conditional closure for complete terminal coordinates

Let Z be a centred finite jointly Gaussian canonical vector with covariance V and Poisson matrix Ω, satisfying $V + i\kappa\Omega \succeq 0$. Write the remaining canonical body and complete record as $X = LZ$, $R = CZ$, omitting irrelevant affine means. Suppose

$$L\Omega L^{\mathsf{T}} = \Omega_X, \qquad L\Omega C^{\mathsf{T}} = 0, \qquad C\Omega C^{\mathsf{T}} = 0. \tag{54}$$

These conditions hold after any affine symplectic evolution when X is a remaining subsystem and R consists of coordinate readings of distinct terminal memory pairs. The last condition means their classical Poisson brackets vanish, also called an isotropic record. This statement uses Poisson geometry; it assumes no quantum operator commutator. General simultaneous classical records need not satisfy (54).

Proposition 7 (terminal-memory posterior closure). Under these hypotheses,

$$\Sigma_{X|R} = LVL^{\mathsf{T}} - LVC^{\mathsf{T}}(CVC^{\mathsf{T}})^{\dagger}CVL^{\mathsf{T}}, \qquad \Sigma_{X|R} + i\kappa\Omega_X \succeq 0. \tag{55}$$

For one body pair this gives $\det \Sigma_{X|R} \geq \kappa^2$. No independent measurement seed or readout product law is assumed.

Proof. Push the initial positive Hermitian matrix through (L, C). By (54) its blocks are

$$\begin{pmatrix} LVL^{\mathsf{T}} + i\kappa\Omega_X & LVC^{\mathsf{T}} \\ CVL^{\mathsf{T}} & CVC^{\mathsf{T}} \end{pmatrix} \succeq 0.$$

The Schur complement proves (55). For a singular record covariance, its null directions have zero cross covariance with X; restrict to its range, or use the displayed pseudoinverse. The one-pair Hermitian determinant is $\det \Sigma - \kappa^2$, so positivity gives the area bound. □

The theorem assumes the full joint covariance restriction and the stated terminal storage geometry. Marginal one-pair bounds do not imply it. **Gaussian-seed corollary (Proposition 3).** A Gaussian measurement seed $\Gamma + i\kappa\Omega_A \succeq 0$ on a discarded subsystem gives the same Schur proof: complex conjugation supplies $\Gamma - i\kappa\Omega_A \succeq 0$, so adding it to the apparatus block of $V + i\kappa\Omega$ leaves the body block $A + i\kappa\Omega_S$ and lower block $B + \Gamma$. Their Schur complement preserves the restriction. Adaptive uses require this condition branch by branch and the declared discard/retained-pointer rule. This is the familiar Gaussian quantum covariance/instrument restriction when $\kappa = \hbar/2$; it is not forced by a bath spectrum.

For a coarse linear record $R_c = BR$, isotropy and body commutation survive, and the Gaussian conditional-variance decomposition gives $\mathrm{Cov}(X \mid R_c) \succeq \mathrm{Cov}(X \mid R)$. This compares the same terminal body; it does not identify two protocols whose additional physical copies have changed that body.

Monitoring a kick must include its physical memory

In the product resonant example, recording the first pointer with independent imprecision variance s^2 and a conjugate kick variance d^2 gives the candidate law

$$s^2 d^2 \geq \kappa^2. \tag{20}$$

If a second pointer infers the reusable momentum $\Pi + D$ exactly, Gaussian conditioning yields

$$\mathrm{Var}(Q \mid R) = \frac{\kappa(\kappa + s^2)}{\kappa + s^2 + G^2\kappa}, \qquad \mathrm{Var}(P - G\Pi \mid R) = \kappa + \frac{G^2\kappa d^2}{\kappa + d^2}. \tag{21}$$

Their product is exactly κ^2 at $s^2 d^2 = \kappa^2$. This is a special case of the conditional-noise determinant below, not a law forced by the bath. The following physical-memory construction realizes the condition from its supplied memory preparation.

Retain the conditional-noise calculation's product resonant body–pointer preparation and first pulse. The body is $X = (Q, P - G\Pi_1)$ and pointer coordinate is $Y_1' = Y_1 + GQ$. Introduce independent memory $A = (Z_A, \Lambda_A)$ with diagonal Gaussian variances $a_0, b_0 > 0$ and $a_0 b_0 = \kappa^2$. A unit pulse with integrated Hamiltonian generator $Y_1'\Lambda_A$ copies the pointer into its coordinate:

$$R_1 = Z_A + Y_1', \qquad K = \Pi_1 - \Lambda_A. \tag{50}$$

All expressions here use the initial random variables. The body stays unchanged: $\{P - G\Pi_1, Y_1'\} = -G + G = 0$. Thus the first imprecision and reusable-pointer kick are $N = Z_A$, $D = -\Lambda_A$, satisfying (20) mechanically from the memory's prior. The copying pulse is a symplectic map, not a passive assignment of an externally stored number.

Now introduce independent monitor memory $B = (Z_B, \Lambda_B)$ with diagonal variances $r^2, d^2 > 0$ and $r^2 d^2 \geq \kappa^2$. A unit pulse generated by $\Lambda_A \Lambda_B$ changes both memory coordinates. The complete terminal readings are

$$R_1' = Z_A + Y_1' + \Lambda_B, \qquad S = Z_B + \Lambda_A. \tag{51}$$

The pointer momentum K is unchanged; grant the same exact second-pointer inference of K as in Proposition 5. The effective noise is $N' = Z_A + \Lambda_B$, $D = -\Lambda_A$. Conditional on S,

$$\begin{aligned} a &= a_0 + d^2, \qquad b = \frac{b_0 r^2}{b_0 + r^2}, \qquad c_{ND} = 0, \\ \det E_S &= \kappa^2 + \frac{b_0 (r^2 d^2 - \kappa^2)}{b_0 + r^2} \geq \kappa^2. \end{aligned} \tag{52}$$

The diagonal product preparation makes N' independent of D, S; the remaining variance is the scalar Gaussian conditioning formula. Equality holds when the monitor memory is also saturated, $r^2 d^2 = \kappa^2$. Equation (32) then gives body posterior area exactly κ for every first gain. Finite second-pointer precision increases it. A sharper monitor $r \downarrow 0$ requires $d^2 \geq \kappa^2 / r^2$ and increasingly disturbs the stored first coordinate. At an uninformative monitor $r \to \infty$ with saturated d^2, its extra disturbance tends to zero. These are finite-model limits; no cutoff removal is implied.

The causal distinction from (34) is measurable in the Poisson brackets:

$$\{R_1', S\} = 1 - 1 = 0, \qquad \{R_1, S\} = 1, \qquad \{R_1', K\} = 1 - 1 = 0. \tag{53}$$

The old numerical value R_1 and the later monitor reading are not the two terminal coordinates of this closed memory construction. If one grants a passive external read of R_1 before the monitor and retains that number without another physical memory coupling, the old escape of (34) returns. Copying it to another physical memory adds that memory's variables and conjugate kick to the calculation. The monitor's disturbance prices an observation of the earlier kick by changing the stored imprecision; it differs from the innovation replacement test's prescribed fresh momentum kick but satisfies the same conditional determinant test. All variances in (52) have action units in the canonical scaling.

For any jointly Gaussian body X and complete record $\mathcal{R}$, the exact answer is the supported Schur complement

$$\mathrm{Cov}(X \mid \mathcal{R}) = K_{XX} - K_{X\mathcal{R}} K_{\mathcal{R}\mathcal{R}}^{\dagger} K_{\mathcal{R}X}. \tag{30}$$

Appending a record decreases this covariance in the positive-semidefinite order. A prior noise covariance therefore needs a conditioning estimate, including indirect observations, before it can enforce a floor.

Here is an exact extension of (20)–(21). Start with the algebraic resonant example $V = \kappa I_6$, $\kappa > 0$, in canonical scaled coordinates. The first pulse still gives body $X = (Q, P - G\Pi)$ and pointer coordinate $Y + GQ$. Storing the latter produces error N and kicks the reusable pointer by D. Grant exact inference of its later momentum $K = \Pi + D$ from the second pointer, as in the ideal second-readout limit. Let (N, D, S) be jointly Gaussian and independent of the six initial body–pointer variables; S is any side record of the readout noise. It can be a monitor read earlier or later, or stored in a separate memory. The complete record is

$$\mathcal{R} = (Y + GQ + N,\ K,\ S).$$

Subtract conditional means given S and put

$$\begin{aligned} E_S &= \mathrm{Cov}\big((N, D)^{\mathsf{T}} \mid S\big) = \begin{pmatrix} a & c_{ND} \\ c_{ND} & b \end{pmatrix} \succeq 0, \\ x &= G^2, \quad u = \kappa + a, \quad v = \kappa + b, \\ \mathcal{D} &= (\kappa x + u) v - c_{ND}^2 > 0. \end{aligned} \tag{31}$$

The symbol c_{ND} is a covariance, distinct from the speed c. All of a, b, c_{ND}, u, v have action units.

Proposition 5 (the conditional recording law, necessary and sufficient in this model). The exact posterior determinant is

$$\boxed{\det \mathrm{Cov}(X \mid \mathcal{R}) = \kappa^2 \left[1 + \frac{x(ab - c_{ND}^2 - \kappa^2)}{\mathcal{D}}\right].} \tag{32}$$

For every nonzero gain G, the posterior area is at least κ if and only if $\det E_S \geq \kappa^2$. A finite-precision second reading can only increase the posterior covariance; the exact-reading limit gives the sharp necessity test.

Proof. Condition first on S. The remaining two readings have covariance $B_R = \left(\begin{smallmatrix} \kappa x+u & c_{ND} \\ c_{ND} & v \end{smallmatrix}\right)$, while $A_X = \mathrm{diag}(\kappa, \kappa(1+x))$ and $C_{XR} = \mathrm{diag}(G\kappa, -G\kappa)$. In particular (30) gives

$$\begin{aligned} \Sigma_{11} &= \kappa(uv - c_{ND}^2)/\mathcal{D}, \\ \Sigma_{12} &= -x\kappa^2 c_{ND}/\mathcal{D}, \\ \Sigma_{22} &= \kappa(1+x) - x\kappa^2(\kappa x + u)/\mathcal{D}. \end{aligned} \tag{33}$$

For a compact determinant check, the covariance of the readings conditional on X is $\left(\begin{smallmatrix} u & c_{ND} \\ c_{ND} & b+\kappa/(1+x) \end{smallmatrix}\right)$. Taking the joint determinant in both block orders yields

$$\det \Sigma = \kappa^2 \frac{\kappa u + (1+x)(ub - c_{ND}^2)}{(\kappa x + u)v - c_{ND}^2}.$$

The numerator minus the denominator is $x(ab - c_{ND}^2 - \kappa^2)$, proving (32). The initial strictly positive κI_6 makes the reading covariance invertible, including singular E_S. □

A concrete indirect-monitor test. Let N, D be independent with $\mathrm{Var}\, N = a_0 > 0$, $\mathrm{Var}\, D = b_0 > 0$ and $a_0 b_0 = \kappa^2$, so the marginal recording law (20) is saturated. Allow the side record $S = D + M$, where M is an independent Gaussian monitor error of variance $r^2 > 0$. Then

$$a = a_0, \qquad b = \frac{b_0 r^2}{b_0 + r^2}, \qquad c_{ND} = 0, \qquad \det E_S = \kappa^2 \frac{r^2}{b_0 + r^2} < \kappa^2. \tag{34}$$

Every informative monitor, even a noisy one, makes (32) strictly smaller than κ^2 for $G \neq 0$. At $r \downarrow 0$, the kick is known and $\det \Sigma = \kappa^2 u/(u + \kappa G^2) \to 0$ as $G \to \infty$. At $r \to \infty$ the original saturated repair is recovered. All limits keep their order and apparatus assumptions explicit. A realizable monitor may itself disturb the pointer; the passive monitor in this test is a granted operation whose physical admissibility is exactly what a closure mechanism must settle.

For arbitrary correlated readout noise with covariance E_0 and side record covariance V_S, the matrix to test is

$$E_S = E_0 - \mathrm{Cov}((N, D), S) V_S^{\dagger} \mathrm{Cov}(S, (N, D)). \tag{35}$$

The common bath's spectrum can set E_0, but it leaves this conditional innovation to be controlled. In the general correlated prior of (8), use (30) on the full joint record; (32)'s iff is asserted for its stated product-prior example only. Theorems 1–2 already cover arbitrary finite body–pointer cross correlations. Proposition 3 supplies a sufficient composition law on every retained subsystem; (34) identifies a side observation that must be priced or excluded by that law.

Complete-record information and adaptive likelihoods

Fix one body, $m > 0$, a window $[0, \tau]$ and a family $\mathfrak{A}$ of finite, non-adaptive affine apparatus protocols with Gaussian preparations and noises independent of the force hypothesis. Each protocol includes its entire retained memory, bath monitors, reused pointers and terminal readings in one record R_π. Linear dynamics gives

$$R_\pi = \mu_\pi + B_\pi \theta + F v_\pi + \xi_\pi, \qquad \xi_\pi \sim N(0, V_\pi), \tag{26}$$

where θ comprises the unknown initial position and velocity. For preparation-ignorant tests take a full maximal invariant $Q_\pi R_\pi$, with $Q_\pi B_\pi = 0$. Put $b_\pi = Q_\pi v_\pi$, $W_\pi = Q_\pi V_\pi Q_\pi^{\mathsf{T}}$ and

$$I_\pi = b_\pi^{\mathsf{T}} W_\pi^{\dagger} b_\pi \quad \text{if } b_\pi \in \operatorname{Ran} W_\pi; \qquad I_\pi = \infty \text{ otherwise.} \tag{27}$$

The finite value is the force Fisher information. The singular case has a noiseless force-sensitive direction. The projection retains all invariant information, including correlations with side records; it is not a choice to discard inconvenient memory. With known preparation take $Q_\pi = I$ instead.

Proposition 4 (exact uniform-continuity criterion). In this family, for equal prior probabilities the best invariant error is

$$P_\pi(F) = \Phi(-|F|\sqrt{I_\pi}/2). \tag{28}$$

For $F \neq 0$ the $I_\pi = \infty$ value is zero; at $F = 0$ it is $1/2$. Writing $I_* = \sup_{\pi \in \mathfrak{A}} I_\pi$, the complete-record maximal-invariant laws $P_{\pi,F}$ are uniformly continuous at zero in total variation exactly when $I_* < \infty$. The quantifier covers every admitted schedule, number of marks and retained memory:

$$\sup_{\pi \in \mathfrak{A}} \|P_{\pi,F} - P_{\pi,0}\|_{\mathrm{TV}} = 2\Phi(|F|\sqrt{I_*}/2) - 1. \tag{29}$$

For $I_* = \infty$ the right side is one at every $F \neq 0$.

Proof. On the supported subspace whiten the Gaussian noise. The likelihood ratio depends only on the projection along the whitened mean difference, whose length is $|F|\sqrt{I_\pi}$. Thresholding at its midpoint gives (28) and the total variation in (29). Different affine supports in the singular case are disjoint. Monotonicity and continuity of Φ permit the supremum, whether or not attained. Consequently finite I_* gives the common modulus $|F|\sqrt{I_*}/\sqrt{2\pi}$, while infinite I_* gives a jump. □

For the unrestricted independent-mark family of Proposition L, $I_* = \tau^3/(24m\kappa)$ at $\kappa > 0$. Its zero branch has $I_* = \infty$. Thus that proposition uses **uniform continuity over the family**, equivalently continuity of its optimal verdict, rather than pointwise continuity of finite noisy apparatus. A bounded classical family may have $I_* < \infty$ at $\kappa = 0$. For example three passive, independent position readings of variance σ^2 at $0, \tau/2, \tau$, with zero recoil, give the invariant statistic $R_0 - 2R_{\tau/2} + R_\tau$, mean $F\tau^2/(4m)$ and variance $6\sigma^2$. Its information is

$$I = \frac{\tau^4}{96m^2\sigma^2}.$$

More generally a finite total precision budget $\sum_j \delta_j^{-2} \leq C$ gives $I_* \leq C\tau^4/(4m^2)$ even with known preparation. Such budgets add under composition; maintaining the same C requires a resource restriction. Unlimited repeats on the same zero-recoil trajectory make $I_* \to \infty$.

Therefore (29) proves the exact statistical requirement of the continuity route. Supplying it independently for a physically specified family closed under arbitrary finite composition remains necessary. It alone supplies neither a terminal imprecision–kick law nor an action constant common to different m, τ and apparatus classes. The quantity $\tau^3/(24mI_*)$ has action units when $0 < I_* < \infty$; identifying it with a universal mark scale uses the additional mark-model saturation and universality premises.

Fix a protocol with finitely many readings $R_1, \ldots, R_n$, each a finite-dimensional real vector. Include all random seeds, retained memory, noise monitors and terminal readings. Conditional on the previous complete history $r_{<j}$, suppose

$$R_j \mid (r_{<j}, F) \sim N\big(\mu_j(r_{<j}) + F b_j(r_{<j}),\ W_j(r_{<j})\big), \qquad W_j \succ 0. \tag{39}$$

Independent randomized controls may use any force-independent law; retain their values in the initial history, so their common density cancels in (41). Count identical copies of an already

stored reading once; deterministic memory functions remain recoverable from the complete history. The protocol may choose its next gains, time and apparatus from that history. At a fixed history those choices are the same under both force hypotheses, and W_j, μ_j, b_j are independent of F. For a linear Gaussian body–apparatus model with a declared Gaussian preparation, this follows by successive Gaussian conditioning: once the realized gains are fixed, the conditional mean is affine in F and the covariance is independent of F. The complete joint record can be non-Gaussian because the gains depend on previous readings. Preparation-ignorant adaptive tests require their own justified invariant kernels; the non-adaptive information criterion's non-adaptive projection alone does not establish (39) for them.

Define the nonnegative information along a branch by

$$I_j(r_{<j}) = b_j^{\mathsf{T}} W_j^{-1} b_j, \qquad J(r) = \sum_{j=1}^{n} I_j(r_{<j}). \tag{40}$$

Both J and its cap C have inverse force squared units, so $F^2 J$ is dimensionless. Let $p_F(r)$ be the normalized complete-record density obtained by multiplying the conditional kernels. Its Hellinger affinity with p_0 is $\mathcal{A}(F,0) = \int \sqrt{p_F p_0}\, dr$.

Proposition 6 (adaptive affinity and continuity). Exactly,

$$\sqrt{p_F(r) p_0(r)} = p_{F/2}(r) e^{-F^2 J(r)/8}, \qquad \mathcal{A}(F,0) = E_{F/2} e^{-F^2 J(R)/8}. \tag{41}$$

For a family of these protocols, uniform complete-record continuity at $F = 0$ is equivalent to

$$\sup_{\pi} E_{\pi, F/2}\left[1 - e^{-F^2 J_\pi(R)/8}\right] \longrightarrow 0. \tag{42}$$

A sufficient condition, uniform in every admitted schedule, history and retained memory, is $J_\pi(r) \le C < \infty$. It gives

$$\|P_{\pi,F} - P_{\pi,0}\|_{\mathrm{TV}} \le \sqrt{1 - e^{-F^2 C/4}} \le \tfrac{1}{2} |F| \sqrt{C}. \tag{43}$$

This deterministic budget is sufficient; rare high-information branches need only satisfy the exact averaged condition (42).

Proof. At the same history, completing the Gaussian square gives

$$\sqrt{g_j(r_j; F) g_j(r_j; 0)} = g_j(r_j; F/2) e^{-F^2 I_j(r_{<j})/8}.$$

Multiply these identities, then integrate, proving (41). For any two normalized densities with affinity $\mathcal{A}$,

$$1 - \mathcal{A} \le \|P - Q\|_{\mathrm{TV}} \le \sqrt{1 - \mathcal{A}^2}.$$

The lower bound follows from $\min(p,q) \le \sqrt{pq}$. For the upper bound apply Cauchy–Schwarz to $|p - q| = |\sqrt{p} - \sqrt{q}|(\sqrt{p} + \sqrt{q})$ and divide its integral by two. These inequalities prove (42); (41) and $J \le C$ prove (43), using $1 - e^{-u} \le u$. □

Information also composes exactly. The score is the sum of $b_j^{\mathsf{T}} W_j^{-1}(R_j - \mu_j - F b_j)$. Each summand has zero conditional mean and conditional variance I_j. When $E_F J < \infty$, orthogonality of these martingale differences gives full-record Fisher information

$$\mathcal{I}(F) = E_F J(R). \tag{44}$$

Joining two blocks of readings adds their branchwise J values, with the second block conditioned on the complete first history. Budgets therefore add; a fixed cap C is not preserved under unrestricted repetition merely because each apparatus has a finite cap. For a non-adaptive Gaussian record J is deterministic, recovering the non-adaptive information criterion's criterion.

A noiseless force-sensitive conditional direction violates the finite-budget hypothesis; it cannot be discarded as irrelevant memory.

An indirect observation can carry force information even when its marginal law is force-independent. For example let $R = Fv + N$, $S = N + M$, with independent centred Gaussian N, M of variances $n^2, r^2 > 0$. In record order R, S, the first kernel has $I_1 = v^2/n^2$, and $S \mid (R, F) \sim N(R - Fv, r^2)$ has $I_2 = v^2/r^2$. Consequently

$$J = v^2(n^{-2} + r^{-2}). \tag{45}$$

This agrees with conditioning the noise on S, whose variance is $n^2r^2/(n^2 + r^2)$. The raw fact that S has no force-dependent mean does not make it harmless after retaining R.

There is also a quantitative repair of the conditional-noise calculation's two-pointer monitor test. Start with its saturated independent error and kick, $a_0b_0 = \kappa^2$, and side reading $S = D+M$ with independent monitor error M of variance r^2. After this monitor, apply a fresh centred Gaussian kick E to the reusable pointer momentum, independent of the initial variables, N, D, S and of variance e^2. The second pointer now infers $K = \Pi + D + E$. By Proposition 5, for a nonzero first gain the body's conditional area remains at least κ precisely when

$$a_0 \left[\frac{b_0 r^2}{b_0 + r^2} + e^2 \right] \geq \kappa^2, \quad \text{equivalently} \quad e^2 \geq \frac{b_0^2}{b_0 + r^2}. \tag{46}$$

The required fresh variance is exactly the part of D learned by the monitor, $\mathrm{Var}(E[D \mid S])$. At an uninformative monitor $r \to \infty$ its cost tends to zero; at an exact monitor $r \downarrow 0$ it must replace all b_0. Equality restores the determinant to κ^2 for every gain in the stated product-prior model. All these variances have action units in the canonical scaling; (46)'s product has action squared units.

This is a useful conditional construction, with the extra kick openly supplied. A later admissible side record of E must be included and would reduce its innovation again. A monitor dynamics that necessarily creates inaccessible fresh noise of at least (46), including its own retained memory, is the physical mechanism to derive. Adding noise by prescription proves neither that mechanism nor $\kappa > 0$.

For two record constructions at the same force, suppose their conditional kernels have the same $W_j(r_{<j}) \succ 0$ and means $m_{\pi,j}(r_{<j})$, $m_{\sigma,j}(r_{<j})$. Put $\delta_j = m_{\pi,j} - m_{\sigma,j}$ and

$$D_{\pi\sigma}(r) = \sum_j \delta_j^{\mathsf{T}} W_j^{-1} \delta_j. \tag{47}$$

Completing the same square as in (41) gives affinity $E_{\mathrm{mid}} e^{-D_{\pi\sigma}/8}$, where the normalized midpoint law uses the average conditional means and the common conditional covariances. In particular, if every branch has $D_{\pi\sigma} \leq \epsilon_{\pi\sigma}^2$, then

$$\|P_\pi - P_\sigma\|_{\mathrm{TV}} \leq \sqrt{1 - e^{-\epsilon_{\pi\sigma}^2/4}} \leq \epsilon_{\pi\sigma}/2. \tag{48}$$

This is an adaptive complete-record refinement criterion. Covariances that change under refinement need an additional comparison; equality of the kernels' covariance is a stated hypothesis here.

For an explicit mechanical input use the harmonic end-kick result in the refinement note, Proposition 3b (local full-read). Fix z_0, w, ρ, τ as there. Suppose at every common history the conditional mean discrepancy is $\delta_j = H_j(r_{<j})[z_\pi(t_j) - z_\sigma(t_j)]$, with the same adaptive sampling time t_j under both constructions. Define $\ell_j = \|W_j^{-1/2} H_j\|_{\rho \to 2}$ and require the branchwise budget $\sum_j \ell_j^2 \leq L^2$, uniform in schedules and memory. Equations (6f) there and (47)–(48) then prove

$$\|P_\pi - P_\sigma\|_{\mathrm{TV}} \leq \frac{L}{2}\, w e^{\rho\tau} (\tau + \rho^{-1})(|\pi| + |\sigma|) \|z_0\|_\rho. \tag{49}$$

Every constant is explicit. The estimate permits arbitrarily many readings when their total whitened sensitivity obeys the declared budget. If the noise widths shrink so fast that this budget diverges, the mechanical Cauchy bound supplies no statistical convergence, as the exact constant-force example (38) already demonstrates. The same-noise linear readout model gives a concrete instance; a general back-reacting body–apparatus model must prove the assumed conditional mean comparison and covariance control.

The finite common-deflection cancellation and partition residuals give two statistical tests (source declarations at newtonlean commit `381d5e9`, read in full; written Gaussian consequences, not newly compiled formal proofs). First, if two observed motions receive the same acceleration history, differencing them removes that random history too. More generally if a retained signal is $r + L\eta$, with bath covariance C_η, a difference map D_r leaves covariance

$$D_r L C_\eta L^{\mathsf{T}} D_r^{\mathsf{T}}. \tag{36}$$

If $D_r L = 0$ it is zero regardless of the bath amplitude. This is a modern noise-model consequence of the finite cancellation identity, not a historical assertion about radiation. A noise floor must survive accessible relative observations as well as the conditional subtraction (35). Common driving alone does not guarantee that survival.

Second, PartitionComparison.partition_gap and PartialCell.candidate_partial_residual and the partial-cell proof separate the actual end-kick polygon from its rational constant-force comparison map. At any rational sample time $t \le \tau$ the exact position residual is $a_{\mathrm{acc}} c_\pi(t)$, where

$$c_\pi(t) = \tfrac{1}{2}\left(\sum_{j \text{ complete}} h_j^2 + u^2\right), \qquad 0 \le c_\pi(t) \le \tfrac{1}{2} t|\pi|, \qquad a_{\mathrm{acc}} = F/m. \tag{37}$$

Here u is the partial final duration and $|\pi|$ bounds every complete and partial cell. The formal theorem uses represented fractions and cross-multiplication equivalence; (37) is its rational-value reading. It fixes the force signal's geometric discrepancy uniformly over partitions. It supplies no measurement law.

For a single Gaussian position record of width σ_π the total variation between polygon and corrected-map records is exactly

$$2\Phi\left(\frac{|F| c_\pi(t)}{2 m \sigma_\pi}\right) - 1. \tag{38}$$

Thus the mechanically vanishing residual gives statistically convergent records only when its noise-normalized size vanishes. A width $\sigma_\pi = F_0 c_\pi(t)/m$ with fixed force $F_0 > 0$ leaves (38) constant for every nonzero F, even as the mesh vanishes. A fixed positive width makes it tend to zero. For the complete multivariate record the corresponding condition is $\|W_\pi^{-1/2} Q_\pi (v_\pi - v_{\mathrm{cont}})\| \to 0$ on its supported subspace; a noiseless residual direction again has total variation one. This distinguishes uniform mechanical refinement from uniform complete-record refinement by an explicit calculation.

Physical Gaussian composition and retained relative modes

Keep the complete-record precision schedule's fixed, force-independent partition and jointly Gaussian product preparation. The body is centred initially, with $A, B > 0$ and $AB \ge \kappa^2$. Allow $\kappa = 0$ in the classical formulas. Define $Y = \sum_i h_i R_i$ at the terminal cut. Partial sums of this expression are bookkeeping for projections of the physical memory coordinates; every copy pulse is still part of the dynamics. This definition grants no extra apparatus for passively storing an earlier numerical read.

During one cell of duration h, drift under the deterministic force, apply its right-endpoint

copy, and add that memory coordinate with weight h. For $X = (Q, P, Y)^{\mathsf{T}}$ the exact update is

$$\begin{aligned} X' &= M_h X + F f_h + \zeta_h, \\ M_h &= \begin{pmatrix} 1 & h/m & 0 \\ 0 & 1 & 0 \\ h & h^2/m & 1 \end{pmatrix}, \qquad f_h = \begin{pmatrix} h^2/(2m) \\ h \\ h^3/(2m) \end{pmatrix}, \\ \zeta_h &= (0, -\Lambda, hZ)^{\mathsf{T}}, \qquad N_h = \operatorname{Cov} \zeta_h = \operatorname{diag}(0, \kappa^2 r h, h/r). \end{aligned} \tag{65}$$

The new pair is independent of all earlier variables. Thus a block's Gaussian channel is described by (M, f, N), and successive blocks obey

$$\begin{aligned} (M_2, f_2, N_2) \circ (M_1, f_1, N_1) &= (M_2 M_1,\ M_2 f_1 + f_2,\ M_2 N_1 M_2^{\mathsf{T}} + N_2), \\ N_{321} &= M_3 M_2 N_1 M_2^{\mathsf{T}} M_3^{\mathsf{T}} + M_3 N_2 M_3^{\mathsf{T}} + N_3. \end{aligned} \tag{66}$$

Multiplication and the displayed noise sum give exactly the same M, f, N in both three-cell parenthesizations. In particular, each earlier momentum kick propagates through the later position and record updates. The accumulated record of any subblock has weight equal to its duration when converted to an average, so its terminal projection also associates. These descriptors specify the projected law of the declared physical copy schedule. Their composite generally differs from the single-cell descriptor (M_H, f_H, N_H) at the total duration: the composite retains the daughter timing and noise data. Arbitrary triples are not asserted to have a one-memory Hamiltonian realization.

For a block of duration H, put $\omega_i = h_i/H$ and define

$$\begin{aligned} s &= \sum_i \omega_i t_i, & k_i &= H - t_i, \\ b_j &= \sum_{i>j} \omega_i (t_i - t_j), & \Gamma &= \sum_j \omega_j b_j, \\ \sigma_t^2 &= \sum_i \omega_i t_i^2 - s^2, & Z_c &= \sum_i \omega_i Z_i, \qquad L_c = \sum_i \Lambda_i. \end{aligned}$$

The pair (Z_c, L_c) is canonical, saturated and Gaussian, with variances $((rH)^{-1}, \kappa^2 r H)$. Resolve every initial momentum as $\Lambda_i = \omega_i L_c + \Lambda_i^{\perp}$. Then

$$\operatorname{Cov}(\Lambda_i^{\perp}, \Lambda_j^{\perp}) = \kappa^2 r H (\omega_i \delta_{ij} - \omega_i \omega_j). \tag{67}$$

Its covariance with L_c vanishes, and all memory momenta are independent of Z_c and the initial body. Joint Gaussianity therefore makes the relative vector independent of the entire coarse vector. It is degenerate in the direction $\sum_i \Lambda_i^{\perp} = 0$; no inverse on that null direction is used.

As in the two-cell comparator, shear the effective memory to $Z_c - \Gamma L_c/m$, copy at s, and drift to H. These controls depend only on the partition. Let $(Q_c, P_c, R_{\mathrm{co}})$ be this physical comparator. Summing (63) gives the exact coupling

$$\begin{aligned} (Q_f, P_f, Y/H) &= (Q_c, P_c, R_{\mathrm{co}}) - (U/m, 0, V/m) + (0, 0, F\sigma_t^2/(2m)), \\ U &= \sum_i k_i \Lambda_i^{\perp}, \qquad V = \sum_i b_i \Lambda_i^{\perp}, \\ \operatorname{Cov}(U, V) &= \kappa^2 r H \begin{pmatrix} \operatorname{Var}_\omega k & \operatorname{Cov}_\omega(k, b) \\ \operatorname{Cov}_\omega(k, b) & \operatorname{Var}_\omega b \end{pmatrix}. \end{aligned} \tag{68}$$

Here $\operatorname{Var}_\omega$ denotes the finite weighted variance with weights ω_i. The deterministic difference is the force response of the physical schedule, not a force-dependent correction which the unknown-force apparatus is assumed able to apply.

For three equal cells of length ℓ,

$$\begin{aligned} k &= (2\ell, \ell, 0), & b &= (\ell, \ell/3, 0), \\ s &= 2\ell, & \Gamma &= 4\ell/9, \qquad \sigma_t^2 = 2\ell^2/3, \\ \operatorname{Cov}(U,V) &= \kappa^2 r\ell^3 \begin{pmatrix} 2 & 1 \\ 1 & 14/27 \end{pmatrix}, \quad \det \operatorname{Cov}(U,V) = \kappa^4 r^2 \ell^6/27. \end{aligned} \tag{69}$$

For example $\operatorname{Var}_\omega k = 2\ell^2/3$, $\operatorname{Cov}_\omega(k,b) = \ell^2/3$, and $\operatorname{Var}_\omega b = 14\ell^2/81$. Multiplication by $\kappa^2 r H = 3\kappa^2 r\ell$ yields (69).

At $\kappa > 0$ this independent residue has rank two in the $(Q, Y/H)$ plane. Even if the deterministic means are matched, the prescribed saturated effective pair plus one independent scalar entering additively along a fixed vector supplies a covariance increment of rank at most one. It cannot reproduce (69). This excludes that specific truncated rule; correlated re-preparations and other physical coarse apparatus have not been excluded. A nonlinear vector-valued function of one random scalar can have covariance rank two and is also outside this exclusion. Equations (65)–(66) retain the full necessary timing, force-response and noise data. At $\kappa = 0$ the residue vanishes and the timing-dependent mean difference remains.

Fix one body of mass $m > 0$ with canonical position and momentum (q, p), drifting under $H = p^2/(2m) - Fq$ for a deterministic force F. Its centred initial Gaussian has diagonal variances $A, B > 0$, $AB \geq \kappa^2$. Let $r > 0$ be a supplied precision per unit time. For $u, v > 0$, $u + v = h$, prepare two independent memory pairs (Z_i, Λ_i), independent of the body, with $\operatorname{Var} Z_i = 1/(rh_i)$, $\operatorname{Var} \Lambda_i = \kappa^2 r h_i$, $h_1 = u$, $h_2 = v$, and zero coordinate–momentum covariance. Their coordinates have position units and their momenta have momentum units. This satisfies the joint condition (22) for $\kappa > 0$. The same classical formulas also make sense at $\kappa = 0$ with zero memory kicks.

At times u and h apply unit symplectic copy pulses generated by $q\Lambda_1$ and $q\Lambda_2$, respectively. Apart from these pulses the memories are frozen. Immediately after the second pulse, the body and all terminal memory coordinates are exactly

$$\begin{aligned} Q_f &= q + hp/m + Fh^2/(2m) - v\Lambda_1/m, \\ P_f &= p + Fh - \Lambda_1 - \Lambda_2, \\ R_1 &= q + up/m + Fu^2/(2m) + Z_1, \\ R_2 &= q + hp/m + Fh^2/(2m) - v\Lambda_1/m + Z_2. \end{aligned} \tag{57}$$

These are physical terminal coordinates, not passive earlier reads. For example $\{R_1, Q_f\} = v/m - v/m = 0$ and $\{R_1, P_f\} = 1 - 1 = 0$. Likewise $\{R_2, Q_f\} = \{R_2, P_f\} = \{R_1, R_2\} = 0$ and $\{Q_f, P_f\} = 1$. The later copy leaves the first stored coordinate unchanged because it already commutes with the evolved body. Thus (54)–(55) apply to the complete fine record (R_1, R_2).

Put $\alpha = u/h$, $\beta = v/h$, and form the block $R_c = \alpha R_1 + \beta R_2$. The initial memories admit the exact canonical change of variables

$$\begin{aligned} Z_c &= \alpha Z_1 + \beta Z_2, & L_c &= \Lambda_1 + \Lambda_2, \\ Z_{\mathrm{rel}} &= Z_1 - Z_2, & L_{\mathrm{rel}} &= \beta\Lambda_1 - \alpha\Lambda_2, \\ (\operatorname{Var} Z_c, \operatorname{Var} L_c) &= ((rh)^{-1}, \kappa^2 rh), \\ (\operatorname{Var} Z_{\mathrm{rel}}, \operatorname{Var} L_{\mathrm{rel}}) &= (h/(ruv), \kappa^2 ruv/h). \end{aligned} \tag{58}$$

Both pairs have bracket one, all cross brackets and cross covariances vanish, and the pairs are Gaussian independent. In particular $\Lambda_1 = \alpha L_c + L_{\mathrm{rel}}$. At one copy time the relative mode would decouple; the distinct times in (57) leave a real residue.

Set $s = h - \alpha v \in (u, h)$ and $c = \alpha\beta v/m$. The coarse comparator prepares the effective pair (Z_c, L_c), applies the symplectic memory shear $Z_c \mapsto Z_c - cL_c$, copies the body position into that

memory at s, and drifts the body to h. Its terminal variables are

$$\begin{aligned} Q_c &= q + hp/m + Fh^2/(2m) - \alpha v L_c/m, \\ P_c &= p + Fh - L_c, \\ R_{\mathrm{co}} &= q + sp/m + Fs^2/(2m) + Z_c - cL_c. \end{aligned} \tag{59}$$

The copy time and shear are predetermined by the split and do not depend on the unknown force. Both descriptions use the coordinate and momentum of the same body at the same final time h. Their distributions differ because their physical interventions differ; no cross-protocol posterior monotonicity is inferred from (56).

The identities $\alpha u + \beta h = s$ and $\alpha u^2 + \beta h^2 - s^2 = \alpha\beta v^2$ give the exact joint coupling

$$\begin{aligned} (Q_f, P_f, R_c) &= (Q_c, P_c, R_{\mathrm{co}}) - \frac{v}{m} L_{\mathrm{rel}}(1, 0, \beta) + \delta(0, 0, 1), \\ \delta &= \frac{F\alpha\beta v^2}{2m}, \qquad \nu = \frac{\kappa^2 r u v^3}{hm^2}, \qquad w = (1, 0, \beta), \\ V_f &= V_c + \nu w w^{\mathsf{T}}. \end{aligned} \tag{60}$$

The added noise is independent of the entire coarse vector. Forgetting the relative record has retained its kick in the projected law; setting that kick to zero would remove the $\nu w w^{\mathsf{T}}$ term. The floor (55) holds for each physical protocol and for the fine block R_c. The law comparison below is joint, rather than a uniform posterior comparison on every exceptional record.

The coarse covariance V_c is positive definite. Indeed, the invertible transformation $T(Q, P, R) = (Q - hP/m, P, R - Q + \alpha v P/m)$ gives

$$T V_c T^{\mathsf{T}} = \mathrm{diag}(A, B, (rh)^{-1}) + \kappa^2 r h\, a a^{\mathsf{T}}, \qquad a = (s/m, -1, -c), \qquad Tw = (1, 0, -\alpha).$$

Consequently, for $x = \nu w^{\mathsf{T}} V_c^{-1} w$ and $y = \delta^2 (V_c^{-1})_{33}$,

$$x \le \nu(A^{-1} + \alpha^2 r h), \qquad y \le r h \delta^2.$$

Completing the Gaussian square gives the exact affinity between the two joint laws, with $d = \delta(0, 0, 1)$,

$$\mathcal{A} = \frac{(1 + x)^{1/4}}{(1 + x/2)^{1/2}} \exp\left[-\frac{1}{8} d^{\mathsf{T}} (V_c + \nu w w^{\mathsf{T}}/2)^{-1} d \right]. \tag{61}$$

This follows by whitening V_c, rotating its rank-one perturbation to one coordinate, and integrating the square root of the two Gaussian densities. The inverse in the exponent is bounded above by V_c^{-1}. Moreover

$$1 - \frac{\sqrt{1 + x}}{1 + x/2} = \frac{x^2}{4(1 + x/2)(1 + x/2 + \sqrt{1 + x})} \le x^2/8.$$

Thus $1 - \mathcal{A}^2 \le x^2/8 + y/4$, and the adaptive affinity inequality gives $\mathrm{TV} \le \frac{1}{2}\sqrt{x^2/2 + y}$. Since $\max \alpha\beta^3 = 27/256$ and $\max \alpha^3\beta^3 = 1/64$, define

$$\begin{aligned} X_h &= \frac{\kappa^2 r}{m^2} \left[\frac{27 h^3}{256 A} + \frac{r h^4}{64} \right], \\ Y_h &= \frac{729 r F^2 h^5}{262144 m^2}, \\ \| \mathcal{L}(Q_f, P_f, R_c) - \mathcal{L}(Q_c, P_c, R_{\mathrm{co}}) \|_{\mathrm{TV}} &\le \frac{1}{2} \sqrt{X_h^2/2 + Y_h}. \end{aligned} \tag{62}$$

This is $O(h^{5/2})$ for bounded F and fixed A, m, r, κ, and $O(h^3)$ at $F = 0$, uniformly over the split. The split maxima follow by differentiating $\alpha(1 - \alpha)^3$ and by $\alpha\beta \le 1/4$. Here r has units

$(\text{length}^2\text{time})^{-1}$, ν has position-variance units, δ has position units, and X_h, Y_h are dimensionless. As either daughter length tends to zero, the residual in (60) vanishes. At $\kappa = 0$ only the force-dependent mean defect remains.

The comparator's time and shear depend on the split. Equation (62) does not yet define an associative apparatus rule over many depths. Summing it over cells would also require a uniform local preparation bound and a valid conditional comparison after earlier records.

For any fixed, force-independent partition $0 = t_0 < \cdots < t_n = \tau$, $h_i = t_i - t_{i-1}$, use independent fresh memories with variances $(1/(rh_i), \kappa^2 rh_i)$ and unit copy pulses at the right endpoints. Keep every physical terminal memory coordinate. The same calculation as (57) gives

$$\begin{aligned} R_i &= q + t_i p/m + F t_i^2/(2m) - \frac{1}{m}\sum_{j<i}(t_i - t_j)\Lambda_j + Z_i, \\ V_R &\succeq \operatorname{diag}((rh_i)^{-1}), \\ \mathcal{I}_R &\le \frac{r}{4m^2}\sum_i h_i t_i^4 \le \frac{r\tau^5}{4m^2}. \end{aligned} \tag{63}$$

The first formula includes every earlier copy kick. Each Z_i is independent of the body and all initial momenta, so the covariance lower bound follows by adding their diagonal noise covariance. The record covariance is force-independent and its mean derivative is $t_i^2/(2m)$; inverse ordering proves the information bound. Later copies leave earlier stored coordinates unchanged, as in the two-copy construction. The terminal vector therefore satisfies (54), regardless of the number of cells. Its complete force sensitivity obeys, for all F, F',

$$\|P_F^R - P_{F'}^R\|_{\mathrm{TV}} \le \frac{|F - F'|}{2}\sqrt{\mathcal{I}_R} \le \frac{|F - F'|\sqrt{r}\,\tau^{5/2}}{4m}. \tag{64}$$

The first bound is the Gaussian affinity calculation with equal covariances. It includes the whole terminal coordinate record and every function of that record, so relative coordinate channels have not been omitted. Additional apparatus revealing initial Z_i, reuse of their conjugates, or adaptive timing require a new calculation.

The total initial precision is fixed by construction: $\sum_i rh_i = r\tau$. Hamiltonian mechanics does not select this preparation schedule or a universal upper bound on r. Within the stated family, terminal storage and a uniform record-continuity budget are derived, and the posterior floor follows from the positive joint preparation scale. Equation (64) remains true at $\kappa = 0$ with the same imprecision law. Thus continuity in this resource class does not force positive action. Letting r grow without bound leaves the uniform class and is a separate physical-resource question.

A quantitative projected terminal limit

Fix τ, m, r, A, B, κ and the deterministic force F. The partition may be arbitrary; set $\delta = \max_i h_i$. Put $s_j = \tau - t_j$ and

$$\begin{aligned} S_1 &= \sum_i h_i t_i = \frac{\tau^2}{2} + \frac{1}{2}\sum_i h_i^2, \\ S_2 &= \sum_i h_i t_i^2, \qquad 0 \le S_2 - \frac{\tau^3}{3} \le \tau^2\delta, \\ B_j &= \sum_{i>j} h_i(t_i - t_j) = \frac{s_j^2}{2} + e_j, \qquad e_j = \frac{1}{2}\sum_{i>j} h_i^2 \le \frac{\delta s_j}{2}. \end{aligned} \tag{70}$$

For the last identity apply the S_1 identity to the subpartition after t_j. For S_2, the right-endpoint sum exceeds the polynomial area $\tau^3/3$; on each cell its excess is at most τh_i^2, since the slope of

t^2 is at most 2τ. Summing gives the bound. These are finite partition identities and elementary polynomial estimates.

Repeated composition in (66), starting with $Y = 0$, is consequently

$$\begin{aligned} M_\pi &= \begin{pmatrix} 1 & \tau/m & 0 \\ 0 & 1 & 0 \\ \tau & S_1/m & 1 \end{pmatrix}, & f_\pi &= (\tau^2/(2m), \tau, S_2/(2m))^{\mathsf{T}}, \\ N_\pi &= \kappa^2 r \sum_j h_j v_j v_j^{\mathsf{T}} + \operatorname{diag}(0, 0, \tau/r), & v_j &= (-s_j/m, -1, -B_j/m)^{\mathsf{T}}. \end{aligned} \tag{71}$$

Define a candidate three-dimensional Gaussian law P_* by replacing S_1, S_2 in M_π, f_π by $\tau^2/2, \tau^3/3$, and setting

$$N_* = \kappa^2 r \begin{pmatrix} \tau^3/(3m^2) & \tau^2/(2m) & \tau^4/(8m^2) \\ \tau^2/(2m) & \tau & \tau^3/(6m) \\ \tau^4/(8m^2) & \tau^3/(6m) & \tau^5/(20m^2) \end{pmatrix} + \operatorname{diag}(0, 0, \tau/r). \tag{72}$$

The first matrix is the polynomial Gram matrix of $(-s/m, -1, -s^2/(2m))$ over $0 \le s \le \tau$. Thus it is nonnegative. Together with the independent initial body covariance and the last term, it gives a positive definite terminal covariance. This defines a finite terminal Gaussian channel without assuming a limiting mechanical curve or a stochastic process at all times.

Proposition 8 (projected terminal-channel refinement). Let $P_\pi = \mathcal{L}(Q_f, P_f, Y)$ for the physical schedule. Define

$$\begin{aligned} a &= \frac{\tau}{m\sqrt{A}}, & b &= \frac{1}{\sqrt{B}}, & c &= \sqrt{\frac{r}{\tau}}\frac{\tau^2}{2m}, \\ L &= \sqrt{a^2 + b^2 + c^2}, & J &= \sqrt{a^2 + 4c^2}, \\ \beta &= \sqrt{\frac{rB}{\tau}}\frac{\tau\delta}{2m}, & d &= c\delta/\tau, \\ \varepsilon(\delta) &= 2\beta + \beta^2 + \kappa^2 r\left[LJ\delta + \tau(2Ld + d^2)\right], & z(\delta) &= |F|c\delta. \end{aligned}$$

Whenever $\varepsilon(\delta) \le 1/2$,

$$\|P_\pi - P_*\|_{\mathrm{TV}} \le \left[\frac{3}{4}\varepsilon(\delta)^2 + \frac{1}{3}z(\delta)^2\right]^{1/2}. \tag{73}$$

All constants are independent of the number of cells and their relative lengths. The bound is $O(\delta)$ at fixed parameters, uniformly over bounded F. Two partitions are Cauchy by the triangle inequality, with the sum of their bounds (73). This proves refinement independence of the body and the single weighted terminal projection in this apparatus family.

Proof. Apply the invertible, dimensionless coordinate change

$$W = \left(\frac{Q - \tau P/m}{\sqrt{A}}, \frac{P}{\sqrt{B}}, \sqrt{\frac{r}{\tau}}\left[Y - \tau Q + \frac{\tau^2 P}{2m}\right]\right)^{\mathsf{T}}.$$

For the candidate, the initial body and summed coordinate noise give covariance I, and the memory momenta add a nonnegative matrix. Thus $\widehat{V}_* \succeq I$. For a finite partition, the former contribution is $(I + \beta_\pi E_{32})(I + \beta_\pi E_{32})^{\mathsf{T}}$, where $\beta_\pi = \sqrt{rB/\tau}(S_1 - \tau^2/2)/m \le \beta$. Its difference from I has norm at most $2\beta + \beta^2$.

A kick at t_j transforms to

$$w(t_j) + (0, 0, -\sqrt{r/\tau}\, e_j/m)^{\mathsf{T}}, \qquad w(t) = \left(\frac{t}{m\sqrt{A}}, -\frac{1}{\sqrt{B}}, -\sqrt{\frac{r}{\tau}}\frac{t^2}{2m}\right)^{\mathsf{T}}.$$

On $[0,\tau]$, $\|w\| \le L$, $\|w'\| \le J/\tau$, and the error vector has norm at most d. The norm of the derivative of ww^{T} is at most $2LJ/\tau$. Its right-endpoint quadrature error on cell i is therefore at most LJh_i^2/τ. Summing gives at most $LJ\delta$. The error vector changes each outer product by at most $2Ld + d^2$. Equation (71) now proves $\|\widehat{V}_\pi - \widehat{V}_*\| \le \varepsilon(\delta)$. The only mean difference in these coordinates is $F\sqrt{r/\tau}(S_2 - \tau^3/3)/(2m)$ in the third component; (70) bounds its norm by $z(\delta)$.

Whiten further by $\widehat{V}_*^{-1/2}$. Its norm is at most one. The relative covariance eigenvalues are $1 + x_j$, $|x_j| \le \varepsilon \le 1/2$, and the mean difference has norm at most z. The squared Gaussian affinity is

$$\mathcal{A}^2 = \prod_{j=1}^{3} \frac{\sqrt{1+x_j}}{1+x_j/2} \exp\left[-\frac{1}{4}\mu^{\mathsf{T}}(I + E/2)^{-1}\mu\right].$$

For $|x| \le 1/2$, rationalizing the square root gives

$$1 - \frac{\sqrt{1+x}}{1+x/2} = \frac{x^2}{4(1+x/2)(1+x/2+\sqrt{1+x})} \le x^2/4.$$

The denominator is at least $17/4 > 4$, using $1 + x/2 \ge 3/4$ and $\sqrt{1+x} \ge 2/3$. Also $I + E/2 \succeq (3/4)I$, so the exponential's deficit is at most $z^2/3$. The deficit of a product of factors in $[0,1]$ is bounded by the sum of their deficits. Hence $1 - \mathcal{A}^2 \le 3\varepsilon^2/4 + z^2/3$. The affinity inequality of the adaptive likelihood calculation gives (73). □

Here $[r] = (\text{length}^2\text{time})^{-1}$, $[\kappa]$ is action, and Y has length-times-time units. The constants a, b, c, L, J, d have inverse-momentum units; β, ε, z are dimensionless. Every term in (71)–(72) has the covariance units of its coordinate pair. The transformed variables and the statistical bound are dimensionless.

For $\kappa > 0$, each finite P_π satisfies terminal posterior closure (55), since Y is a linear projection of commuting physical terminal coordinates. Its record variance is at least τ/r; the conditional covariance is therefore continuous at P_*, and the limit retains $\det \Sigma_{(Q,P)|Y} \ge \kappa^2$. At $\kappa = 0$, (65)–(73) still hold with all momentum kicks zero; convergence and continuity in this fixed-precision family select no positive action scale.

Physical adaptive feedback and a continuous record

Retain the complete-record precision schedule's force-independent partition and product Gaussian preparation, including the centred initial body. For each pulse choose a force-independent C^2 function $g_i(R_1, \dots, R_{i-1})$, with $|g_i| \le G < \infty$ and bounded first derivatives for each finite protocol. The gains are dimensionless. At t_i use the integrated Hamiltonian generator $\mathcal{G}_i = g_i(R_{<i})Q_i\Lambda_i$. The arguments are the actual stored coordinates of earlier memories, which commute with the current body. This is a supplied nonlinear mechanical coupling, with its full canonical recoil included.

During this pulse Q_i, $R_{<i}$ and the fresh momentum Λ_i stay fixed. Its exact flow is

$$\begin{aligned} P &\longmapsto P - g_i\Lambda_i, \qquad Z_i \longmapsto R_i = Z_i + g_iQ_i, \\ \Pi_j &\longmapsto \Pi_j - (\partial_j g_i)Q_i\Lambda_i \quad (j < i). \end{aligned} \tag{74}$$

Here Π_j denotes the current conjugate of stored coordinate R_j. The symbol Λ_j in all subsequent terminal formulas denotes its initial, fresh pre-copy momentum, rather than its later recoiled conjugate. Old memories are otherwise frozen; their conjugates remain in the full mechanical state, and are isolated from later body couplings and readouts. Later generators use their coordinates directly. Thus the controller uses physical storage throughout. The C^2 regularity makes the recoil-inclusive flow a differentiable canonical map; bounded first derivatives and gains keep its finite-protocol moments finite. No implementation resource bound for such couplings is derived here.

Summing the body impulses gives exactly

$$\begin{aligned} R_i &= Z_i + g_i(R_{<i}) \left[q + t_i p/m + F t_i^2/(2m) - \frac{1}{m} \sum_{j<i} (t_i - t_j) g_j(R_{<j}) \Lambda_j \right], \\ Q_\tau &= q + \tau p/m + F\tau^2/(2m) - \frac{1}{m} \sum_j (\tau - t_j) g_j(R_{<j}) \Lambda_j, \\ P_\tau &= p + F\tau - \sum_j g_j(R_{<j}) \Lambda_j. \end{aligned} \tag{75}$$

Every delivered kick survives. The complete declared coordinate record is $R = (R_1, \dots, R_n)$; old conjugate readouts and terminal-body readouts would require additional measuring couplings and a new law.

Proposition 9 (physical adaptive-gain closure). Under these apparatus assumptions, conditionally on each complete terminal value $R = a$, the body law is Gaussian and satisfies

$$\Sigma_{(Q_\tau, P_\tau)|R=a} + i\kappa \Omega_{\text{body}} \succeq 0. \tag{76}$$

In particular its determinant is at least κ^2 at $\kappa > 0$. Let P_F^R be the adaptive coordinate-record law. Then

$$\begin{aligned} J(a) &\le C = \frac{r G^2 \tau^5}{4m^2}, \\ \mathcal{A}(F, F') &= E_{(F+F')/2} e^{-(F-F')^2 J(R)/8}, \\ \|P_F^R - P_{F'}^R\|_{\text{TV}} &\le \sqrt{1 - e^{-(F-F')^2 C/4}} \le \frac{|F - F'| G \sqrt{r}\, \tau^{5/2}}{4m}. \end{aligned} \tag{77}$$

The cap is uniform in the finite partition and all controllers in the bounded-gain class. The continuity calculation remains valid at $\kappa = 0$.

Proof. At fixed a, freeze the numerical gains at $g_i = g_i(a_{<i})$. Write C_g, L_g for the ordinary linear record and terminal-body matrices of that predetermined-gain copy protocol, and put $c_{g,i} = g_i t_i^2/(2m)$, $V_g = \text{Cov}(C_g Z_{\text{init}})$. Each record has its own independent Z_i with coefficient one, so $V_g \succeq D = \text{diag}((r h_i)^{-1}) \succ 0$.

To check the adaptive density rather than assume Gaussianity, fix the latent variables $U = (q, p, \Lambda_1, \dots, \Lambda_n)$. The map $(Z_i) \mapsto (R_i)$ in (75) is triangular with diagonal derivatives one. Its inverse is $Z_i = a_i - g_i(a_{<i}) Q_i(U; g_{<i}(a), F)$. If φ_i is the initial density of Z_i, integrate against the Gaussian latent law μ_U:

$$p_F(a) = \int \prod_i \varphi_i(a_i - g_i(a_{<i}) Q_i(U; g_{<i}(a), F))\, d\mu_U(U) = \varphi_{F c_g, V_g}(a)\big|_{g=g(a)}. \tag{78}$$

This is the frozen-gain Gaussian density evaluated at the adaptive path. Its derivation establishes normalization and includes all gain dependence; no gain-derivative Jacobian is present. The full adaptive law is generally non-Gaussian.

The same integral before integrating U identifies its conditional law at a as precisely the frozen protocol's Gaussian posterior. At that fixed record the terminal body in (75) is the same affine function $L_g Z_{\text{init}}$, with its force mean. For the auxiliary frozen linear protocol, copies and drifts give (54) for L_g, C_g. Proposition 7 therefore proves (76). This applies its covariance theorem to the frozen protocol; the full nonlinear terminal state need not satisfy a Gaussian covariance closure claim.

At the same path a, $g(a)$ and V_g are identical under both force hypotheses. Completing the Gaussian square in (78) gives

$$\sqrt{p_F(a) p_{F'}(a)} = p_{(F+F')/2}(a) e^{-(F-F')^2 J(a)/8}, \qquad J(a) = c_g^{\mathsf{T}} V_g^{-1} c_g.$$

Inverse ordering and the gain bound imply

$$J(a) \le c_g^{\mathsf{T}} D^{-1} c_g = \frac{r}{4m^2} \sum_i h_i g_i (a_{<i})^2 t_i^4 \le C.$$

Integration proves the affinity identity and lower bound $\mathcal{A} \ge e^{-(F-F')^2 C/8}$. The affinity inequality and $1 - e^{-x} \le x$ give (77). □

Frozen-protocol Gaussian innovations along the observed path are also the actual adaptive conditional kernels along that path. Their information sums to $J(a)$, consistent with (41)–(44). Conditional kernels compose and their branchwise information budgets add. When composing blocks, retain the body's conditional mean, covariance and inherited force sensitivity: applying a fresh local-duration τ^5_{block} cap would drop force information already in the incoming body. Equation (77) is the valid global-duration bound. It also controls every measurable function of the declared full record.

Fixed r, G are supplied resource parameters; (77) selects no positive action. The inserted-cell comparison above treats a prescribed policy depending on the physical weighted past record. Adaptive timing and global propagation under refinement remain open. The initial positive preparation law, universality and radiation calibration remain separate obligations.

Prescribe one force-independent function $g : \mathbb{R} \to \mathbb{R}$ for every partition, with $g \in C^2$, $|g| \le G$, $|g'| \le L$, $|g''| \le D_2$. The argument has the units of Y, length times time; L, D_2 have its inverse and inverse-square units. At the right endpoint of cell i use gain

$$g_i = g(Y_{i-1}), \qquad Y_{i-1} = \sum_{j<i} h_j R_j.$$

This is a function of the physical frozen coordinates, not a numerical record supplied to an external controller. The pulse is (74); each old conjugate receives $-h_j g'(Y_{i-1}) Q_i \Lambda_i$. The new first daughter's conjugate also recoils during the second pulse. All are retained and isolated as specified in the adaptive-gain construction. Preparation and times remain force-independent.

Proposition 10 (physical inserted-cell defect). Condition analytically on an incoming mechanical state with body (q, p) and weighted memory projection y. Compare a parent cell $h = a + b$ to two daughter cells $a, b > 0$, using the same incoming state and force. Prepare the daughter pairs independently as before and couple the coarse pair by

$$Z_c = (aZ_1 + bZ_2)/h, \qquad \Lambda_c = \Lambda_1 + \Lambda_2.$$

It is canonical and saturated, with variances $((rh)^{-1}, \kappa^2 rh)$. It has a canonical independent relative pair

$$Z_{\mathrm{rel}} = Z_1 - Z_2, \qquad \Lambda_{\mathrm{rel}} = (b\Lambda_1 - a\Lambda_2)/h.$$

Their cross brackets and covariances vanish. The fine apparatus retains both pairs; the coarse experiment may leave the relative pair as a spectator. This coupling does not delete any fine kick.

Put

$$\begin{aligned} q_1 &= q + ap/m + Fa^2/(2m), & q_h &= q + hp/m + Fh^2/(2m), \\ g_0 &= g(y), & \eta &= ag_0 q_1 + aZ_1, \\ g_1 &= g(y + \eta), & d_g &= g_1 - g_0. \end{aligned}$$

Fine minus coarse terminal values satisfy exactly

$$\begin{aligned} \Delta Q &= -bg_0 \Lambda_1/m, \\ \Delta P &= -d_g \Lambda_2, \\ \Delta Y &= -\frac{abg_0}{m}\left[p + \frac{F(a+h)}{2}\right] + bq_h d_g - \frac{b^2 g_0 g_1 \Lambda_1}{m}. \end{aligned} \tag{79}$$

Here the common terminal record projection is Y: a weight h on the coarse reading, and weights a, b on the daughters. Conditioning on q, p, y grants the physical controller no readout of q or p. Its only argument remains the stored coordinate projection.

Write E_s for the conditional expectation over fresh pairs and set

$$V_s = L^2(a^2G^2q_1^2 + a/r).$$

Then

$$\begin{aligned} E_s\Delta Q = E_s\Delta P = 0, \qquad & E_s d_g^2 \le V_s, \\ E_s(\Delta Q)^2 &\le \kappa^2 r a b^2 G^2/m^2, \\ E_s(\Delta P)^2 &\le \kappa^2 r b V_s, \\ \operatorname{Var}_s \Delta Y &\le b^2 q_h^2 V_s + \kappa^2 r a b^4 G^4/m^2. \end{aligned} \tag{80}$$

Its only bias obeys

$$|E_s\Delta Y| \le \frac{abG}{m}(|p| + |F|h) + b|q_h| \left[LaG|q_1| + \frac{D_2}{2}(a^2G^2q_1^2 + a/r) \right]. \tag{81}$$

There is no singular factor in either daughter duration.

Proof. In the fine experiment the first position is q_1 and the second is $q_h - bg_0\Lambda_1/m$. The second gain is g_1, because the first physical reading increments y by η. The two readings are $Z_1 + g_0q_1$ and $Z_2 + g_1(q_h - bg_0\Lambda_1/m)$. Subtract the coarse reading $Z_c + g_0q_h$ and use $q_1 - q_h = -b[p + F(a+h)/2]/m$ to obtain (79).

η depends only on the fresh coordinate Z_1 and is independent of both fresh momenta. Its second moment is $a^2g_0^2q_1^2 + a/r$. The Lipschitz bound gives $E_sd_g^2 \le V_s$, and independence gives the first two variances. The two random terms in ΔY have zero covariance: condition on Z_1 and use $E_s\Lambda_1 = 0$. In fact its variance is

$$b^2q_h^2 \operatorname{Var}_s d_g + b^4g_0^2E_sg_1^2\,\kappa^2 ra/m^2.$$

This proves (80). Taylor expansion gives $|E_sd_g| \le LaG|q_1| + (D_2/2)(a^2G^2q_1^2 + a/r)$; the momentum term in ΔY has mean zero. Equation (81) follows. □

These are state-dependent conditional bounds. A usable uniform averaging bound follows from the physical protocol itself. Let

$$M_i = \sum_{j \le i} g(Y_{j-1})\Lambda_j.$$

In the filtration generated by the initial body and consumed pointer pairs, this is a martingale. Old conjugates depend only on these variables; retaining their recoil does not reveal a future pointer. For $v_i = \kappa^2 rG^2 \sum_{j \le i} h_j$, conditional Gaussian moments give $EM_i^2 \le v_i$ and $EM_i^4 \le 3v_i^2$. For the latter, the fourth-moment recursion has increments at most $6(v_i - v_{i-1})v_{i-1} + 3(v_i - v_{i-1})^2$. The finite-martingale L^4 maximal inequality therefore yields $\|\max_i |M_i|\|_4 \le (4/3)3^{1/4}|\kappa|G\sqrt{r\tau}$.

The position impulse sum is a finite sum of interval durations times past M_i, bounded by $\tau \max_i |M_i|/m$. Minkowski and the centred initial Gaussian moments consequently give

$$\begin{aligned} K_Q &= 3^{1/4}\left[\sqrt{A} + \frac{\tau\sqrt{B}}{m} + \frac{4|\kappa|G\tau\sqrt{r\tau}}{3m}\right] + \frac{|F|\tau^2}{2m}, \\ K_P &= 3^{1/4}\left[\sqrt{B} + \frac{4|\kappa|G\sqrt{r\tau}}{3}\right] + |F|\tau, \\ \|\sup_{t\le\tau}|Q(t)|\|_4 &\le K_Q, \qquad \|\sup_{t\le\tau}|P(t)|\|_4 \le K_P, \\ \|\max_i |Y_i|\|_4 &\le G\tau K_Q + \frac{4}{3}\,3^{1/4}\sqrt{\tau/r}. \end{aligned} \tag{82}$$

The suprema here are of each finite physical piecewise trajectory; both momentum values at a pulse are included. The last estimate uses the Gaussian martingale $\sum_{j\le i} h_j Z_j$ of variance at most τ/r. All constants are independent of partition size and schedule. The initial preparation assumption is the one in the Gaussian descriptor.

For local defects evaluated along an admitted parent schedule, its unsplit drift continuations q_1, q_h also satisfy this fourth-moment bound. In particular $E(q_h^2 q_1^2) \le K_Q^4$. For disjoint parent cells with $h_i \le \delta$, $\sum_i h_i = \tau$, one has $\sum_i a_i b_i \le \tau\delta/4$. Averaging (80)–(81) gives the local budgets

$$\begin{aligned}
\sum_i E(\Delta P_i)^2 &\le \frac{\tau\delta}{4}[\kappa^2 L^2 + \kappa^2 r L^2 G^2 K_Q^2 \delta],\\
\sum_i E|E_s \Delta Y_i| &\le \frac{\tau\delta}{4} C_Y(\delta),\\
C_Y(\delta) &= \frac{G}{m}(K_P + |F|\delta) + LGK_Q^2 + \frac{D_2}{2}(G^2 K_Q^3 \delta + K_Q/r).
\end{aligned} \tag{83}$$

These quantify centred local noise and accumulated local bias, before propagation. They are not bounds on the difference of two full feedback protocols: later gains also respond to incoming discrepancies.

Every term in (79)–(83) has its coordinate's units. At constant gain $d_g = 0$, the extra momentum defect vanishes. At $\kappa = 0$, $\Delta Q = \Delta P = 0$ but ΔY generally remains random, because the first coordinate noise changes the second gain. At $g \equiv 0$ all defects vanish. The bounds survive very unequal splits and select no positive action scale.

Path construction from arbitrary finite refinements

Retain the fixed apparatus family of the weighted-record policy, including one prescribed g, a fixed duration, and force-independent deterministic partitions. For each partition π, keep the actual continuous body position $Q_\pi(t)$. Linearly interpolate the post-copy momenta at its cuts to give $\overline{P}_\pi$, and the weighted memory coordinates at its cuts to give $\overline{Y}_\pi$. Include initial values $p, 0$. These interpolations are retrospective functions of the finite experiment; they do not give the controller an extra stored reading. Write $\mathsf{X}_\pi = (Q_\pi, \overline{P}_\pi, \overline{Y}_\pi)$.

Proposition 11 (adaptive path refinement). The laws of these continuous paths have a unique limit as $|\pi| \to 0$, independent of deterministic partition schedules. On a countable nested family they can be coupled to converge in probability in the uniform path metric. The constants below are independent of cell count. The limit follows from finite Gaussian refinements; no limiting noise process or mechanical curve is assumed.

For two partitions take their finite common refinement ϱ, with atomic durations ϵ_e. Prepare independent variables Λ_e, ξ_e of variances $\kappa^2 r\epsilon_e, \epsilon_e/r$ and bracket $\{\xi_e, \Lambda_f\} = \epsilon_e \delta_{ef}$. For a parent cell I of length H, its apparatus pair is

$$Z_I = H^{-1} \sum_{e\subset I} \xi_e, \qquad \Lambda_I = \sum_{e\subset I} \Lambda_e. \tag{84}$$

It has exactly the original independent saturated Gaussian preparation within each partition. Canonical completion supplies the relative modes as in the weighted-record policy. They remain in the finer experiment. Equation (84) couples experiments; it grants no simultaneous readout of their different apparatus.

For a proof localization, clip the position appearing in the record drift to $[-R, R]$, keeping the body kick unchanged. This auxiliary recursion is not asserted a mechanical Hamiltonian outside the clipped region. Coupled to the original experiment, it agrees until an original copy position leaves the region. Equation (82) bounds that probability by K_Q^4/R^4. Its body moment bound K_P is unchanged, since it uses only the bounded predictable gains.

Distribute each clipped parent kick over its atoms with the same parent-left gain, and drift an auxiliary position under these atomized kicks. Distribute its record increment as

$\epsilon_e g(Y_{I,\mathrm{left}})\,\mathrm{clip}_R(Q_{I,\mathrm{right}})+\xi_e$. The right position before the parent kick is predicted by $Q_{I,\mathrm{left}}+HP_{I,\mathrm{left}}/m+FH^2/(2m)$, determined entirely at parent left. All these coefficients are predictable in the atomic filtration. The auxiliary momentum and record match the clipped parent recursion at parent endpoints; its position differs by the kick timing. The prediction is analytical and assumes no force-aware apparatus control.

Normalize by $q_0=\sqrt{A}$, $p_0=\sqrt{B}$, $y_0=\sqrt{\tau/r}$, and put

$$\alpha=\frac{p_0}{mq_0},\quad \theta=\frac{q_0}{y_0},\quad \ell=Ly_0,\quad b_0=\frac{\kappa^2 r}{B},\quad \overline{R}=R/q_0.$$

Here α,θ,b_0 have inverse-time units, and $\ell,\overline{R}$ are dimensionless. Let x,v,z denote the normalized auxiliary coordinates. At an atomic left endpoint, their lags from the parent-right position predictor and parent-left record have squared mean bounds, respectively,

$$\begin{aligned}B_x(\delta)&=\frac{4}{m^2A}[\kappa^2 rG^2\tau\delta^2+K_P^2\delta^2+F^2\delta^4/4+\kappa^2 rG^2\delta^3],\\ B_z(\delta)&=(rG^2R^2\delta^2+\delta)/\tau,\qquad \delta=|\pi|.\end{aligned}\tag{85}$$

The first lag is the sum of four terms: completed-parent timing discrepancy; remaining parent drift; remaining force drift; and the current distributed-kick area. Orthogonality of the centred parent increments bounds the first variance by $\kappa^2 rG^2\tau\delta^2/m^2$. The other squared bounds are $K_P^2\delta^2/m^2$, $F^2\delta^4/(4m^2)$ and $\kappa^2 rG^2\delta^3/m^2$. Squaring their sum costs four. For the record lag, its bounded parent drift and fresh coordinate noise have zero cross term, giving B_z directly.

For the two partitions define

$$\begin{aligned}\mathcal{B}_x&=B_x(\delta)+B_x(\delta'),\qquad \mathcal{B}_z=B_z(\delta)+B_z(\delta'),\\ c_R&=\tau\alpha^2+12b_0\ell^2+6\tau\theta^2(G^2+\overline{R}^2\ell^2),\\ E_R&=12b_0\ell^2\tau\mathcal{B}_z+6\tau^2\theta^2(G^2\mathcal{B}_x+\overline{R}^2\ell^2\mathcal{B}_z).\end{aligned}\tag{86}$$

c_R has inverse-time units and E_R is dimensionless. Let D_k be the expectation of the sum of the three componentwise maxima squared of auxiliary differences up to atomic endpoint k. Common coordinate noise cancels in the record difference. Momentum differences are martingale sums. The gain difference has squared expectation at most $3\ell^2(E|\Delta z|^2+\mathcal{B}_z)$. The normalized record drift difference has squared expectation at most

$$6\theta^2[G^2E|\Delta x|^2+\overline{R}^2\ell^2E|\Delta z|^2+G^2\mathcal{B}_x+\overline{R}^2\ell^2\mathcal{B}_z].$$

These follow by inserting the two auxiliary states between the parent predictors; clipping is one-Lipschitz. The martingale maximal inequality gives a factor four for momentum; Cauchy–Schwarz over duration at most τ bounds both drift sums. Thus the finite recurrence and its elementary product bound are

$$D_k\le E_R+c_R\sum_{e\le k}\epsilon_e D_{e-1},\qquad D_n\le E_R\prod_e(1+c_R\epsilon_e)\le E_Re^{c_R\tau}.\tag{87}$$

No stability constant depends on the number of atoms or parents.

It remains to compare auxiliary paths with the displayed physical paths. At completed parent cuts their position timing errors form a martingale with maximal second moment at most $4\kappa^2 rG^2\sum_I H_I^3/m^2$. The unfinished parent's kick area has maximal second moment, summed over parents, at most the same bound. Squaring their sum costs two for each term. This gives the position contribution sixteen in

$$\mathcal{H}(\delta)=\frac{16\kappa^2 rG^2\tau\delta^2}{m^2A}+\frac{49}{\sqrt{27}}\left[\frac{\kappa^2 rG^2}{B}+\frac{1}{\tau}\right]\sqrt{\tau\delta}.\tag{88}$$

For momentum and record the differences are finite Gaussian bridges between a parent increment and its linear interpolation. A partial increment's maximal L^4 norm is at most $(4/3)3^{1/4}$ times the square root of its parent variance. Adding the terminal linear term gives $(7/3)3^{1/4}$ times that root. Its fourth power is $2401/27$ times the variance squared. Sum over parents and use $\sum_I H_I^2 \le \tau\delta$ to obtain $49/\sqrt{27}$ in (88). This bounds the expectation of the sum of componentwise squared uniform errors by $\mathcal{H}$.

Between atomic endpoints, auxiliary record noise jumps cancel in the difference, and body force terms cancel. Its componentwise maxima are consequently controlled by the endpoint maxima in (87). Minkowski, Markov and removal of clipping give, for normalized paths $\widehat{\mathsf{X}} = (Q/q_0, \overline{P}/p_0, \overline{Y}/y_0)$,

$$\Pr\left(\sup_{t\le\tau} |\widehat{\mathsf{X}}_\pi(t) - \widehat{\mathsf{X}}_{\pi'}(t)| > \varepsilon\right) \le \frac{2K_Q^4}{R^4} + \frac{\left[\sqrt{E_R e^{c_R\tau}} + \sqrt{\mathcal{H}(\delta)} + \sqrt{\mathcal{H}(\delta')}\right]^2}{\varepsilon^2}. \tag{89}$$

First fix R and refine both meshes; then let $R \to \infty$. The clipping tail vanishes in this stated order. A rapidly growing c_R therefore causes no missing cell-count uniformity.

For a countable nested family, adjoin independent Gaussian relative pairs at each refinement. This gives a consistent coupling of every finite apparatus preparation. The continuous displayed paths are Cauchy in probability by (89). Completeness of continuous path space in its uniform metric gives a limit: choose a subsequence with summable probability bounds, obtain an almost surely uniformly Cauchy subsequence, and then use the probability-Cauchy bound for the full sequence. No infinite physical readout is postulated in this argument. For arbitrary deterministic partition sequences, the finite common refinement comparison gives convergence to this same **law**. A single universal coupling of all real cut schedules is not required.

The displayed momentum differs from the actual impulsive momentum by at most the largest parent kick. Its fourth-moment sum is at most $3(\kappa^2 r G^2)^2\tau\delta$, so this difference tends to zero in probability. The finite mechanical identity therefore passes to the uniform limit:

$$Q_*(t) = q + \frac{1}{m}\int_0^t P_*(s)\,ds, \qquad Q_* \in C^1, \quad P_* \in C. \tag{90}$$

The finite identity uses ordinary integrals of piecewise polynomials; the limiting integral follows from uniform convergence. Separate evolution laws for P_* and Y_* have not yet been identified here. At $\kappa = 0$, the body is the constant-force parabola conditional on q, p; the stated Gaussian preparation keeps those initial variables random in its unconditional law.

Each finite displayed $\overline{Y}_\pi$ encodes its own complete coordinate record, since $R_i = [\overline{Y}_\pi(t_i) - \overline{Y}_\pi(t_{i-1})]/h_i$. The limiting path does not thereby encode the readings generated by every earlier finite protocol: their kicks and gains change under refinement. Equation (89) is a path-law construction, rather than complete-record TV convergence between those different experiments. Conditioning on the declared limiting record is addressed in the conditional weak-limit theorem.

Nevertheless (77) applies to every finite displayed record path. Weak convergence on this Polish space and lower semicontinuity of total variation yield for the limiting record laws

$$\|P_F^{Y_*} - P_{F'}^{Y_*}\|_{\mathrm{TV}} \le \sqrt{1 - e^{-(F-F')^2 r G^2 \tau^5/(16m^2)}} \le \frac{|F-F'|G\sqrt{r}\,\tau^{5/2}}{4m}. \tag{91}$$

Thus uniform complete-record force continuity survives both the weighted-record path limit and the zero-action branch in this fixed resource family.

Conditioning on the entire limiting record

The limiting posterior need not be Gaussian for its floor to survive. The following covariance statement supplies the needed closure.

Proposition 12 (conditional covariance under a weak limit). Let (X_n, Y_n) converge weakly to (X, Y), with $X_n \in \mathbb{R}^2$ and Y_n in one Polish record space. Suppose $|X_n|^2$ is uniformly integrable and, for one fixed $\kappa \geq 0$,

$$\operatorname{Cov}(X_n \mid Y_n) + i\kappa\Omega \succeq 0 \quad \text{almost surely,} \qquad \Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Then

$$\operatorname{Cov}(X \mid Y) + i\kappa\Omega \succeq 0 \quad \text{almost surely,} \qquad \det \operatorname{Cov}(X \mid Y) \geq \kappa^2. \tag{92}$$

The moment hypothesis is in any fixed coordinate units. The fourth moments (82) provide it for the body variables used here.

Proof. Fix $a \in \mathbb{C}^2$, a bounded continuous complex function f of the record and a bounded nonnegative continuous function w. Conditional decomposition of prediction error gives

$$E[w(Y_n)|a^*X_n - f(Y_n)|^2] + i\kappa a^*\Omega a\, Ew(Y_n) \geq 0. \tag{93}$$

Indeed its conditional squared error is $a^*\Sigma_n a + |a^*\mu_n - f|^2$. The first term together with the real number $i\kappa a^*\Omega a$ is nonnegative by hypothesis. The test in (93) is continuous with quadratic growth in X_n. Uniform integrability passes its expectation through weak convergence, giving the same inequality for (X, Y).

For fixed w, approximate $a^*E[X \mid Y]$ in $L^2(w\, dP_Y)$ by bounded continuous complex functions; these are dense for a finite Borel measure on a Polish space. Minimizing the prediction error in this way gives

$$E[w(Y)\{a^* \operatorname{Cov}(X \mid Y)a + i\kappa a^*\Omega a\}] \geq 0.$$

Testing against all such nonnegative w proves that the integrable quantity in braces is nonnegative almost surely. Equivalently its associated finite signed Borel measure is positive. Use a countable dense set of complex vectors with rational components, then continuity of the quadratic form, to obtain one null set outside which the matrix is positive. Its determinant is $\det \operatorname{Cov}(X \mid Y) - \kappa^2$, proving (92). □

The integrability requirement has content. With dimensionless $\kappa = 1$ and a constant record, let $X_n = 0$ with probability $1 - 1/n$, and give each of $\pm\sqrt{2n}\, e_1, \pm\sqrt{2n}\, e_2$ probability $1/(4n)$. Its covariance is I, but its weak limit is zero. Its fourth moment is $4n$ and its squared norm is not uniformly integrable. Equation (82) prevents this escape of covariance into rare tails in the mechanical construction.

Apply the proposition with terminal body variables $X_n = (Q_{\pi_n}(\tau), P_{\pi_n}(\tau))$ and $Y_n = \overline{Y}_{\pi_n}$. Each finite displayed record generates exactly the same sigma-field as that protocol's full coordinate record. Equation (76) supplies the finite posterior floor; (82) supplies integrability; Proposition 11 supplies joint weak convergence. Thus (92) holds conditional on the **entire declared limiting weighted-record path**. Neither convergence of conditional Gaussian means nor a Gaussian limiting posterior is required.

This closes observation by the stipulated retained coordinates at the supplied scale. Additional conjugate readouts or measuring channels require their own dynamics and record law. At $\kappa = 0$ the theorem gives ordinary covariance positivity. Equations (89)–(93) derive continuum and posterior closure within the supplied preparation family; none selects its positive branch.

Limiting evolution and retained controller conjugates

The sources in the limiting evolution can be constructed from (84), rather than postulating an extra bath. Retain the same fixed policy, preparations and deterministic duration. On a countable nested refinement family, let L_π, N_π linearly interpolate the cumulative fresh increments $\Lambda_i, h_i Z_i$. These are retrospective source descriptions. Their adapted left-step versions have the same limit.

Proposition 13 (evolution from the physical increments). On this coupling there are independent continuous centred Gaussian sources L, N, independent of the initial body, such that

$$\begin{aligned} E[L_s L_t] &= \kappa^2 r \min(s,t), & E[N_s N_t] &= r^{-1}\min(s,t), \\ E\sup_{t\le\tau}|L_\pi(t)-L_t|^2 &\le \frac{49}{\sqrt{27}}\kappa^2 r\sqrt{\tau|\pi|}, & E\sup_{t\le\tau}|N_\pi(t)-N_t|^2 &\le \frac{49}{\sqrt{27}r}\sqrt{\tau|\pi|}. \end{aligned} \tag{94}$$

At $\kappa = 0$, L is identically zero. No division by κ or additional source is needed.

Source construction. Cumulative increments agree at existing cuts by (84). Between two parent cuts, the difference from their linear interpolation is the finite Gaussian bridge estimated in (88). Its fourth moment is at most $2401/27$ times the squared parent variance. Sum over parents, use $\sum h_i^2 \le \tau|\pi|$ and take a square root to obtain (94), initially against a finer partition. Completion and Fatou give the displayed limits. Gaussian finite-dimensional laws, covariances and independence pass to them. In particular their disjoint increments are independent. Continuous path space was completed here, not supplied with a new physical noise.

Use the completed natural filtration of q, p and the histories of L, N through time t. Future apparatus increments are excluded. The interpolations in (94) and the adaptive path limit are generally not adapted inside their cells. Past-step versions and the vanishing bridge errors show that their continuous limits L, N, Q_*, P_*, Y_* are adapted.

For predictable step integrands f, independence of fresh increments gives the isometry and finite maximal bound

$$E\left|\int_0^\tau f_s\,dL_s\right|^2 = \kappa^2 r E\int_0^\tau |f_s|^2 ds, \qquad E\sup_{t\le\tau}\left|\int_0^t f_s\,dL_s\right|^2 \le 4\kappa^2 r E\int_0^\tau |f_s|^2 ds. \tag{95}$$

Completion in this norm defines the integral for the integrands below. The integrals of simple predictable integrands against continuous L have continuous paths; the maximal bound preserves that property in the uniform limit in probability.

Suppress stars on the constructed limit. Its exact evolution is

$$\begin{aligned} Q_t &= q + \frac{1}{m}\int_0^t P_s\,ds, \\ P_t &= p + Ft - \int_0^t g(Y_s)\,dL_s, \\ Y_t &= N_t + \int_0^t g(Y_s) Q_s\,ds. \end{aligned} \tag{96}$$

To pass the finite sums to these equations, on a parent cell use its left record and the pre-copy position predictor $Q_i = Q_{i-1} + h_i P_{i-1}/m + F h_i^2/(2m)$. Both are measurable before its fresh pair is consumed. Uniform path convergence in probability, bounded continuous g, g' and (82) give convergence of $g(Y_{i-1})$ and $g'(Y_{i-1})Q_i$ in $L^2(ds\,dP)$ to their limiting integrands. The fourth moments supply uniform integrability for the products involving Q_i. Equation (95) upgrades convergence of the momentum sums to uniform L^2 convergence. Ordinary drift quadratures give the other two equations; the actual impulsive momentum and its interpolation have the same limit by the adaptive path limit.

Two adapted solutions with the same sources and initial body agree: stop where either position exceeds R, use the no-lag difference estimate behind (87), and apply Gronwall with zero incoming difference. Bounded gains give the same body moment bounds (82) for such solutions, so letting $R \to \infty$ removes the stopping. This proves uniqueness within this supplied source family. The arbitrary-partition law of the adaptive path limit is therefore the law of (96).

Here square brackets denote stochastic quadratic covariation, distinct from a canonical bracket. From (94)–(96),

$$\begin{aligned} [L]_t &= \kappa^2 r t, & [N]_t &= t/r, \quad [L,N]_t = 0, \\ [P]_t &= \kappa^2 r\int_0^t g(Y_s)^2 ds, & [Y]_t &= t/r, \quad [P,Y]_t = 0. \end{aligned} \tag{97}$$

Q has finite variation. Also $Y - N$ has finite variation and $[g(Y), L] = \int g'(Y)\, d[Y, L] = 0$. Thus Itô and Stratonovich give the same momentum integral **in this model**. Other apparatus couplings require their own calculation. Zero instantaneous covariation does not remove the correlations used by the posterior.

The isolated conjugates remain in the limit. Every old memory conjugate has the exact terminal value

$$\Pi_j^{\mathrm{term}} = \Lambda_j - h_j \sum_{i>j} g'(Y_{i-1}) Q_i \Lambda_i.$$

For a fixed continuous dimensionless deterministic test ϕ, reorder the finite sum to obtain

$$\begin{aligned} \sum_j \phi(t_j)\Pi_j^{\mathrm{term}} &= \sum_i \left[\phi(t_i) - \left(\sum_{j<i} h_j \phi(t_j) \right) g'(Y_{i-1}) Q_i \right] \Lambda_i, \\ \Gamma(\phi) &= \int_0^\tau [\phi(s) - H_\phi(s) g'(Y_s) Q_s]\, dL_s, \qquad H_\phi(s) = \int_0^s \phi(t) dt. \end{aligned} \tag{98}$$

The first line converges in L^2 to the second by (95): the inner deterministic Riemann sums converge uniformly, and the remaining predictable factors have the integrability already proved. This is an L^2-valued linear functional on continuous tests, with explicit bound

$$\|\Gamma(\phi)\|_2 \le |\kappa| \sqrt{r}\, \|\phi\|_\infty \left[\sqrt{\tau} + L K_Q \sqrt{\tau^3/3} \right].$$

It retains every recoil without asserting a samplewise finite signed measure or a pointwise conjugate density. Indeed $\mathrm{Var}(\Lambda_j/h_j) = \kappa^2 r/h_j$ diverges under refinement when $\kappa > 0$. The future recoil sum is orthogonal to this consumed increment by the martingale property, so the variance of $\Pi_j^{\mathrm{term}}/h_j$ is at least as large. At $\kappa = 0$, L and Γ both vanish. None of these latent conjugates is added to the declared coordinate record.

Finite filtering mechanism. At a copy, condition on the preceding coordinate record, so its gain is numerical. Let the pre-copy Gaussian body covariance be $\left(\begin{smallmatrix} a & c \\ c & b \end{smallmatrix}\right)$, $D = ab - c^2$, and put $k = r g^2 h$. Fresh independent pointer variances $1/(rh), \kappa^2 rh$ give the exact posterior update

$$a^+ = \frac{a}{1+ka}, \qquad c^+ = \frac{c}{1+ka}, \qquad b^+ = b + \kappa^2 k - \frac{k c^2}{1+ka}. \tag{99}$$

This is the conditional Gaussian covariance formula for $R = Z + gQ$, $P^+ = P - g\Lambda$. Direct multiplication gives

$$D^+ = \frac{D + \kappa^2 ka}{1+ka} = \kappa^2 + \frac{D - \kappa^2}{1+ka}. \tag{100}$$

Free drift preserves D. Thus every finite copy preserves saturation or contracts the excess determinant toward κ^2. This identifies the mechanism inside the admitted affine body-copy family. It does not assume a continuous Gaussian posterior filter.

For each fixed deterministic t, choose refinements containing that cut and apply the conditional weak-limit theorem to the protocol truncated there. Partition independence gives $\det \mathrm{Cov}((Q_t, P_t) \mid Y_{[0,t]}) \ge \kappa^2$. Conditioning a past body on future records is a different question. The momentum source has units of momentum, the record source of length times time, and (95)–(100) preserve these units. With $\kappa = \hbar/2$ the positive scale is still supplied.

Nonlinear copies defeat the classical Gaussian extension

The preceding closure theorem permits nonlinear feedback from old records but copies the current body **linearly**. Test an extension with one nonlinear terminal body copy. This is a conditional test of an enlarged apparatus family, not a new physical premise selecting κ.

Its finite pointer precision and coupling parameters are specified below; it makes no claim for arbitrary preassigned resource caps that exclude them.

Prepare independent centred Gaussians

$$Q \sim N(0, A), \quad P \sim N(0, \kappa^2/A), \quad Z \sim N(0, E), \quad \Lambda \sim N(0, \kappa^2/E), \qquad A, E, \kappa, q_0 > 0.$$

Both canonical pairs saturate the initial phase-covariance restriction. Here q_0 is an apparatus length, and A, E have length-squared units. Supply the integrated classical pulse $\mathcal{G} = f(Q)\Lambda$, $f(q) = q^2/(2q_0)$. Its exact global canonical flow, including all recoil, is

$$Q' = Q, \qquad P' = P - \frac{Q\Lambda}{q_0}, \qquad R = Z + \frac{Q^2}{2q_0}, \qquad \Lambda' = \Lambda. \tag{101}$$

The declared complete terminal coordinate record is the single R; its conjugate is retained and isolated. Both terminal body variables Poisson-commute with R. There is no omitted side record or discarded body impulse.

Proposition 14 (failure of the nonlinear classical extension). If $E \le A^2/(20000q_0^2)$, then on an open interval of terminal readings with positive probability, $\det \mathrm{Cov}((Q', P') \mid R) < \kappa^2$.

Proof. At $R = 0$, the position posterior density is proportional to

$$\exp\left[-\frac{q^2}{2A} - \frac{q^4}{8Eq_0^2}\right].$$

Initial P, Λ remain independent of this position and of each other. Symmetry gives zero means and zero body cross covariance. With $v = E[Q^2 \mid R = 0]$, the conditional determinant is therefore

$$\frac{D_{\mathrm{cl}}}{\kappa^2} = \frac{v}{A} + \frac{v^2}{Eq_0^2} = 2\lambda\mu_2(\lambda) + 8\mu_2(\lambda)^2, \tag{102}$$

where $\ell^4 = 8Eq_0^2$, $\lambda = \ell^2/(2A)$ and $\mu_2(\lambda)$ is the second moment of the density proportional to $e^{-x^4-\lambda x^2}$. Differentiation under its integrals gives $\mu_2'(\lambda) = -\operatorname{Var}_\lambda(x^2) < 0$. At zero, $\mu_2(0) = r_\Gamma = \Gamma(3/4)/\Gamma(1/4)$.

The needed constant can be bounded without numerical evaluation. Gamma recurrence and strict Cauchy–Schwarz give

$$\frac{\Gamma(15/4)}{\Gamma(13/4)} = \frac{77}{15}r_\Gamma, \qquad \Gamma(15/4)^2 < \Gamma(13/4)\Gamma(17/4) = \frac{13}{4}\Gamma(13/4)^2.$$

Thus $r_\Gamma^2 < 2925/23716 < 1/8$ and $2r_\Gamma < 3/4$. The assumed precision gives $\lambda = \sqrt{2E}q_0/A \le 1/100$. Equation (102) now implies

$$\frac{D_{\mathrm{cl}}}{\kappa^2} < \frac{5850}{5929} + \frac{3}{400} = 1 - \frac{13813}{2371600} < 1. \tag{103}$$

The record density is strictly positive near zero. Its conditional moments are continuous there by Gaussian domination, so strict failure persists on an open interval of positive probability. The result does not depend on choosing a conditional version at one null event. □

A bounded C^2 copy also fails. Define a C^1 cutoff $\chi(s) = 1$ for $s \le 1$, $\chi(s) = 0$ for $s \ge 2$, and $\chi(s) = 1 - 3(s-1)^2 + 2(s-1)^3$ between them. Put

$$f_K(q) = \frac{1}{q_0}\int_0^{|q|} s\chi(s/K)ds.$$

This is even and C^2, agrees with the quadratic on $|q| \le K$, and is constant $23K^2/(20q_0)$ beyond $2K$. It satisfies $|f_K'| \le \min(|q|, 2K)/q_0$ and $|f_K''| \le 4/q_0$. The conditional normalizing integral,

Q^2 moment and $(f_K')^2$ moment at $R = 0$ converge to those of the quadratic by dominated convergence against the initial Gaussian. Evenness preserves zero conditional position mean. Hence the corresponding determinant converges to the strict bound (103), and some sufficiently large finite K also violates the floor on an interval. This establishes existence of a bounded copy; it supplies no quantitative threshold for K.

Exact quantum reference and curvature. Now supply $\hbar = 2\kappa$ and normalized real pure Gaussian body and pointer wavefunctions with position variances A, E. The unitary copy $e^{-if(Q)\Lambda/\hbar}$ followed by its coordinate readout gives

$$\psi_a(q) \propto \exp\left[-\frac{q^2}{4A} - \frac{(a - f(q))^2}{4E}\right].$$

Its position density is exactly the classical posterior above. Its conditional momentum statistics are obtained with $P = -i\hbar\partial_q$. Define the score

$$s_a(q) = -\partial_q \log|\psi_a(q)|^2 = \frac{q}{A} - \frac{(a - f(q))f'(q)}{E}.$$

The amplitude is real, so momentum mean and symmetrized body cross covariance vanish. Integration by parts, with vanishing tails, gives

$$\begin{aligned} \mathrm{Var}_{\mathrm{qm}} P &= \kappa^2 E_a[s_a^2] = \kappa^2 E_a[s_a'], \\ \mathrm{Var}_{\mathrm{qm}} P - \mathrm{Var}_{\mathrm{cl}} P' &= -\frac{\kappa^2}{E} E_a[(a - f)f''], \\ \mathrm{Var}_{\mathrm{qm}}(P \mid R = 0) &= \kappa^2\left[\frac{1}{A} + \frac{3v}{2Eq_0^2}\right]. \end{aligned} \tag{104}$$

The last line exceeds the classical value by $\kappa^2 v/(2Eq_0^2)$. Also $E_0[Qs_0] = 1$, so $vE_0[s_0^2] \geq 1$ by Cauchy–Schwarz and the quantum determinant is at least κ^2. For a nonsymmetric copy use $E_a[(Q - E_aQ)s_a] = 1$ instead.

This is the curvature compensation supplied by the specified quantum model. Classical mechanics does not uniquely force that compensation. The Heisenberg recoil formulas agree with (101), while operator-state conditioning and classical probability conditioning give different momentum statistics. Recoil bookkeeping alone cannot decide between them.

A finite witness beyond the covariance. The full conditional quantum state also leaves the positive canonical phase-space class. Use its Wigner representation

$$W(q,p) = \frac{1}{2\pi\hbar}\int e^{-ipu/\hbar}\psi_0(q + u/2)\psi_0(q - u/2)du.$$

At $q = 0$, the normalized Fourier data of the momentum slice are

$$\frac{\psi_0(u/2)\psi_0(-u/2)}{|\psi_0(0)|^2} = e^{-u^2/(8A) - u^4/(128Eq_0^2)}.$$

Differentiation at zero yields the absolutely convergent slice moment

$$\frac{\int p^4 W(0,p)dp}{\int W(0,p)dp} = 3\kappa^4\left[\frac{1}{A^2} - \frac{1}{Eq_0^2}\right] < 0 \quad \text{if } Eq_0^2 < A^2. \tag{105}$$

The denominator is $|\psi_0(0)|^2 > 0$. The state and its Wigner function are Schwartz, so these moments are ordinary integrals. A nonnegative slice cannot have a negative fourth moment. Consequently $W(0, p_*) < 0$ at some p_*; continuity gives an open negative phase-space region. The stricter preparation bound (103) satisfies this condition as well.

Thus this full conditional state's canonical Wigner representation cannot be a positive classical pushforward density. Positive models of record statistics alone, separate marginal distributions or the corrected covariance are not excluded. Other representations and contextual encodings have not been tested here.

Weak nonlinear copies and complete-record sufficiency

Sharp terminal precision was inessential to the nonlinear-copy test's failure. The following canonical pulse family has exact composition at every measurement exposure, including arbitrarily weak copies. There is **no free body drift between the copies**. The parameter T is cumulative measurement exposure in the pointer preparation below; this is a controlled recording model, not a finite-mass free-flight limit. It tests the nonlinear recording extension already proposed.

Fix $A, r, q_0, \kappa > 0$ and a finite partition with $h_i > 0$, $\sum_i h_i = T$. Prepare independent Gaussians

$$\begin{aligned} Q &\sim N(0, A), & P &\sim N(0, \kappa^2/A), \\ Z_i &\sim N(0, (rh_i)^{-1}), & \Lambda_i &\sim N(0, \kappa^2 r h_i). \end{aligned}$$

Every pointer pair separately saturates the supplied covariance floor. Its coordinate variance diverges as $h_i \downarrow 0$, so the fine copies are weak. Apply the exact canonical pulse $\mathcal{G}_i = f(Q)\Lambda_i$, $f(q) = q^2/(2q_0)$, and retain every coordinate reading $R_i = Z_i + f(Q)$. All conjugates remain isolated. The terminal body and coarse pointer variables are

$$\begin{aligned} P_T &= P - f'(Q)L_T, & L_T &= \sum_i \Lambda_i, \\ Y_T &= \sum_i h_i R_i = Tf(Q) + T\bar{Z}, & \bar{Z} &= T^{-1}\sum_i h_i Z_i, \\ \operatorname{Var}\bar{Z} &= (rT)^{-1}, & \operatorname{Var} L_T &= \kappa^2 rT, \qquad \{\bar{Z}, L_T\} = 1. \end{aligned} \tag{110}$$

The coarse pair is independent of the initial body and has independent coordinate and conjugate. Its variance product is again κ^2. The body momentum includes every impulse; it Poisson-commutes with each retained R_i.

Exact complete-record sufficiency. At a fixed record vector,

$$\prod_i p(R_i \mid Q) \;\propto\; \exp\left[rY_T f(Q) - \frac{rT}{2} f(Q)^2\right]. \tag{111}$$

The omitted factor is independent of Q. Every residual coordinate $R_i - Y_T/T = Z_i - \bar{Z}$ is independent of $(Q, P, L_T, \bar{Z})$: its covariance with $\bar{Z}$ is $1/(rT) - 1/(rT) = 0$, and the pointer coordinate law is Gaussian. Thus the entire vector and its sufficient statistic Y_T give exactly the same conditional body law. This is sufficient-statistic compression of a declared complete record, rather than discarding informative data.

The independent source completion of the source and recoil construction gives continuous L, N with variances $\kappa^2 rt, t/r$. Here the limiting law is simply

$$Y_t = tf(Q) + N_t, \qquad P_t = P - f'(Q)L_t. \tag{112}$$

At a fixed $T > 0$, the bridge $Y_t - (t/T)Y_T = N_t - (t/T)N_T$ is independent of (Q, P, L_T, Y_T). Zero Gaussian covariance proves this first for finitely many bridge times; continuity and a countable dense set extend it to the full path. Consequently the entire limiting coordinate record $Y_{[0,T]}$ gives the same posterior as Y_T. Arbitrary partitions, nested or not, have the same terminal law by (110)–(111).

Proposition 15 (refinement-invariant weak-copy obstruction). Put

$$s = \frac{A\sqrt{rT}}{q_0}, \qquad z = \sqrt{r/T}\, Y_T, \qquad x = Q/\sqrt{A}.$$

If $m = E[x^2 \mid z]$, then

$$\begin{aligned} \rho_z(x) \;&\propto\; e^{-(1-sz)x^2/2 - s^2x^4/8}, \\ \frac{\det \operatorname{Cov}((Q, P_T) \mid Y_{[0,T]})}{\kappa^2} &= m + s^2 m^2 \le 1 - \frac{s}{(1+s)^2} < 1 \quad (z \le -1). \end{aligned} \tag{113}$$

The inequality holds for every $T > 0$. For $0 < s \le 1$ its failure event has probability at least

$$c_0 = [2\Phi(1) - 1]\Phi(-3/2) \ge \frac{e^{-5/2}}{2\pi} > 0, \tag{114}$$

uniformly in exposure and subdivision. Here Φ is the standard normal distribution function.

Proof. The posterior in (113) is (111) times the initial Gaussian. It is even. Initial P and L_T are independent of that posterior, so conditional momentum mean and cross covariance vanish, while

$$\mathrm{Var}(P_T \mid z) = \kappa^2/A + \kappa^2 rTAm/q_0^2.$$

Multiplication by $\mathrm{Var}(Q \mid z) = Am$ proves the determinant identity. For $z \le -1$, compare the posterior with the Gaussian of precision $1 - sz \ge 1 + s$. Reweighting by $e^{-s^2x^4/8}$ decreases its second moment: the covariance of x^2 with this decreasing function of x^2 is nonpositive. Thus $m \le (1 - sz)^{-1} \le (1 + s)^{-1}$. Substitute into the increasing function $m + s^2m^2$ to obtain (113).

The actual record is $z = \xi + (s/2)X^2$, with independent standard normal ξ, X. If $s \le 1$, the event $\{|X| \le 1,\ \xi \le -3/2\}$ implies $z \le -1$ and has probability c_0. Finally $2\Phi(1) - 1 \ge 2\varphi(1)$ and $\Phi(-3/2) \ge \frac{1}{2}\varphi(2)$, by integrating the standard normal density φ on $[-1, 1]$ and $[-2, -3/2]$. Their product gives the explicit lower bound in (114). □

The failure approaches the saturated boundary as $T \downarrow 0$; its probability does not approach zero in this model. The determinant deficit is of order $\sqrt{T}$ on a fixed negative-innovation interval, while independent momentum recoil variance is of order T. The next calculation establishes that this mechanism also occurs for bounded copies, without relying on the quadratic tail.

Proposition 16 (bounded curvature test, with constants). Let $f \in C^2(\mathbb{R})$ be bounded, with bounded f', f''. Replace the quadratic above by this copy and use $Q \sim N(\mu, A)$, $P \sim N(0, \kappa^2/A)$. Write

$$\begin{aligned} g &= f - Ef(Q), & |g| &\le M, \qquad M > 0, \\ U &= (Q - \mu)/\sqrt{A}, & t &= M\sqrt{rT}, \\ z &= \sqrt{r/T}\,[Y_T - TEf(Q)], & H &= \frac{\mathrm{Cov}(U^2, f(Q))}{M} = \frac{AEf''(Q)}{M}, \\ C_D &= 80 + \frac{16AE[f'(Q)^2]}{M^2}. \end{aligned} \tag{115}$$

Prior expectations in these constants use the stated Gaussian. Uniformly for $t \le 1/4$ and $|z| \le 2$,

$$\left|\frac{D_T(z)}{\kappa^2} - 1 - tzH\right| \le C_D t^2. \tag{116}$$

If $H \ne 0$ and $0 < t \le \min\{1/4, |H|/(2C_D)\}$, there is a complete-record event of probability at least

$$c_1 = \Phi(7/4) - \Phi(5/4) \ge \frac{e^{-49/32}}{2\sqrt{2\pi}} > 0$$

on which $D_T/\kappa^2 \le 1 - t|H|/2 < 1$.

Proof. Put $G = g/M$. The posterior relative to the prior Gaussian has weight $w = e^{tzG - t^2G^2/2}$. For the stated range,

$$|w - 1 - tzG| \le 6t^2, \qquad Ew \ge 1 - 6t^2 \ge 5/8, \qquad w \le 2.$$

For the remainder, set $v = tzG - t^2G^2/2$. Then $|v| \le 17t/8 \le 17/32$, $e^{|v|} < 2$, and $|e^v - 1 - tzG| \le t^2/2 + v^2e^{|v|}/2 \le (321/64)t^2 < 6t^2$. Also $EG = 0$ and $|H| \le 1$. Dividing the first and second moment numerators by Ew gives

$$|E_zU| \le 7t, \qquad |E_zU^2 - 1 - tzH| \le 30t^2.$$

For the second bound, each remainder is at most $6t^2$; the normalization's product with tzH costs at most $12t^3$, and $(Ew)^{-1} \le 2$. Therefore $|\operatorname{Var}_z U - 1 - tzH| \le 79t^2$.

Independence of P, L_T from the coordinate record still gives zero body cross covariance and

$$\frac{D_T(z)}{\kappa^2} = \operatorname{Var}_z U + \frac{At^2}{M^2} \operatorname{Var}_z U \, E_z[f'(Q)^2].$$

The weight bounds imply $\operatorname{Var}_z U \le 4$ and $E_z[f'^2] \le 4E[f'^2]$, proving (116). Two Gaussian integrations by parts give $\operatorname{Cov}(U^2, f) = AEf''$, proving (115).

Choose z of sign opposite to H, with $1 \le |z| \le 2$. Equation (116) then gives the asserted deficit. Since the actual record is $z = \xi + tG(Q)$, $|tG| \le 1/4$, noise $\xi \in [5/4, 7/4]$ guarantees $z \in [1, 2]$; its negative interval guarantees $z \in [-2, -1]$. Either event has probability c_1, independently of Q. Integrating φ over the positive interval gives its explicit lower bound. □

Thus preserving the supplied floor for **every** Gaussian mean and width in this bounded-copy class requires $Ef''(Q) = 0$ for all μ, A. Letting $A \downarrow 0$, boundedness and continuity of f'' give $f''(\mu) = 0$ everywhere. Such a bounded affine function is constant. Ordinary nonconstant affine copies in the unbounded Gaussian family preserve saturation by (99)–(100). The result is a local curvature obstruction to extending that mechanism, not an assumption that the nonlinear posterior can be restarted as a Gaussian.

At $\kappa = 0$, P and every Λ_i are identically zero in this continuation. The record law persists, the determinant is zero, and the ordinary positivity restriction is satisfied. Divisions by κ in (113), (116) apply only to the supplied positive branch. The weak-copy failure selects neither that branch nor its calibration.

The supplied quantum curvature correction and positivity test

Retain the quadratic copy and supplied $\hbar = 2\kappa > 0$ of the nonlinear-copy test. Let s be a dimensionless pulse strength and $U_s = e^{-isf(Q)\Lambda/\hbar}$. In the coordinate density kernel,

$$\rho_s(q_+, z_+; q_-, z_-) = \rho_0(q_+, z_+ - sf(q_+); q_-, z_- - sf(q_-)).$$

Use $q_\pm = q \pm u/2$, $z_\pm = z \pm v/2$. The exact quadratic identity $f(q \pm u/2) = f(q) \pm f'(q)u/2 + f''u^2/8$ changes pointer centre and relative variables to

$$z' = z - sf(q) - sf''u^2/8, \qquad v' = v - sf'(q)u.$$

The Fourier phase then contains $(p + sf'(q)\lambda)u + \lambda v'$. Since multiplication by u^2 corresponds to $-\hbar^2\partial_p^2$, the full Wigner evolution is exactly

$$\begin{aligned} W_s &= e^{sc_3\partial_p^2\partial_z} W_{\mathrm{cl},s}, \qquad c_3 = \frac{\hbar^2}{8q_0} = \frac{\kappa^2}{2q_0}, \\ W_{\mathrm{cl},s}(q,p,z,\lambda) &= W_0(q, p + sf'(q)\lambda, z - sf(q), \lambda), \\ \partial_s W_s &= \left[f'(q)\lambda\partial_p - f(q)\partial_z + c_3\partial_p^2\partial_z \right] W_s. \end{aligned} \tag{106}$$

Define the exponential by its Fourier multiplier $e^{-isc_3k_p^2k_z}$; no convergence of a derivative Taylor series is asserted. It preserves Schwartz regularity. The classical transport and the correction commute because their coefficients depend only on q, λ. Thus (106) is an exact finite kernel identity, not an asymptotic expansion. There are no further terms for this quadratic copy. All three generator terms are dimensionless per unit pulse strength.

The correction is visible after conditioning. Write $n_s(a) = \int W_s(q, p, a, \lambda)\, dq\, dp\, d\lambda$ for the ordinary terminal record density. At $k_p = 0$ the correction multiplier is one and its first k_p

derivative is zero. Consequently record density, conditional position law, first momentum moment and symmetrized QP moment agree with the classical pushforward. The second derivative gives instead

$$\mathrm{Var}_{\mathrm{qm}}(P \mid R = a) - \mathrm{Var}_{\mathrm{cl}}(P' \mid R = a) = 2sc_3 \frac{n_s'(a)}{n_s(a)}. \tag{107}$$

Equivalently, integration by parts of the correction gives $\int p^2 \partial_p^2 W \, dp = 2 \int W \, dp$; higher p derivatives contribute nothing to this second moment. The Fourier calculation justifies this statement without a series assumption. First momentum means agree, so the same formula holds for variances.

For the Gaussian pointer, $n_s(a) = E_Q[\varphi_E(a - sf(Q))] > 0$ and differentiation gives

$$\frac{n_s'(a)}{n_s(a)} = -E_a[a - sf(Q)]/E.$$

At $s = 1$, (107) is exactly $-\kappa^2 E_a[(a - f)f'']/E$ in (104). At zero reading it equals $\kappa^2 v/(2Eq_0^2)$. Thus the finite kernel derives the missing curvature contribution while leaving the measured record law unchanged. Matching that law alone would miss the distinction.

A positive classical transition family cannot have this generator on the full canonical phase space. The adjoint observable generator is

$$\mathcal{A} = -f'(q)\lambda \partial_p + f(q)\partial_z - c_3 \partial_p^2 \partial_z. \tag{108}$$

Any positive transition kernels K_s, starting at the identity and possessing this pointwise generator on smooth compactly supported tests, satisfy the positive-minimum principle: if $u \geq 0$ and $u(x_*) = 0$, then $K_s u(x_*) \geq 0$ implies $\mathcal{A}u(x_*) \geq 0$.

Take a nonnegative smooth compact cutoff equal to one near a point with $p = z = 0$. Multiply it by $u = (p/p_*)^2 e^{\beta z/z_*} + (z/z_*)^2$, with $p_*, z_*, \beta > 0$ in momentum, length and dimensionless units. At that point u and its first derivatives vanish, while $\partial_p^2 \partial_z u = 2\beta/(p_*^2 z_*)$. Therefore

$$\mathcal{A}u(x_*) = -\frac{2c_3\beta}{p_*^2 z_*} < 0, \tag{109}$$

a contradiction. This excludes a positive classical transition family with (108) on the full canonical phase space. Restricted initial-state models, contextual encodings and matching selected moments are not excluded. There is no contradiction with quantum positivity: a pointwise nonnegative phase-space symbol need not represent a positive operator.

The algebraically continued generator loses this correction at $\kappa = 0$ and becomes classical transport. The Wigner-transform definition itself requires $\hbar > 0$. Neither the obstruction nor the conditional compensation selects the positive action branch from independent physical premises.

Bayes' rule isolates a candidate correction without a quantum operator. It does not identify that correction with physical momentum. In the frozen-copy family, with $Q \sim N(\mu, A)$, the posterior score is

$$\sigma_y(q) = -\partial_q \log \rho_y(q) = \frac{q - \mu}{A} - r[y - Tf(q)]f'(q).$$

Gaussian tails in the bounded and quadratic cases justify integration by parts. If $I(\rho) = \int \rho(\partial_q \log \rho)^2 dq$, then

$$\begin{aligned} I(\rho_y) &= E_y[\sigma_y^2] = E_y[\sigma_y'] \\ &= \frac{1}{A} + rTE_y[f'^2] - rE_y[(y - Tf)f''], \\ \kappa^2 I(\rho_y) - \mathrm{Var}_{\mathrm{cl}}(P_T \mid y) &= -\kappa^2 r E_y[(y - Tf)f'']. \end{aligned} \tag{117}$$

For the quadratic, relative to the prior momentum variance κ^2/A, this correction is $-sz + s^2 m/2$. Its leading term cancels the signed fluctuation in (113). The same expression is the

quantum-reference compensation in (104), now displayed as a posterior score identity. Assigning momentum variance $\kappa^2 I(\rho_y)$ is an additional state rule; the covariance floor alone does not uniquely select it.

For a smooth positive prior with the requisite finite score and tail integrability, the likelihood score has conditional mean zero at fixed Q. Expanding the square therefore gives the exact record average

$$E_Y I(\rho_Y) = I(\rho_0) + rTE_{\rho_0}[f'^2]. \tag{118}$$

In the Gaussian case this equals the average classical momentum variance divided by κ^2. The curvature correction in (117) has mean zero because $Y_T - Tf(Q) = N_T$ is independent of Q. It has both signs when the averaged curvature in (115) is nonzero. Adding another independent centred kick to each existing classical posterior can only increase its variance and cannot implement this signed redistribution on every record.

Composition test. For successive conditionally normalized likelihoods, posterior scores add. Their cross terms vanish on record averaging, giving the corresponding sum of conditional likelihood Fisher informations in (118). This probability identity also holds for an adaptive likelihood after its earlier records are fixed; it makes no assertion about the adaptive apparatus's unobserved conjugates.

Unread records impose an additional exact obligation. Let a finite retained label J select smooth positive densities ρ_j with probabilities w_j, and put $\rho = \sum_j w_j \rho_j$. Then $\sigma_\rho(q) = E[\sigma_J(q) \mid Q = q]$, so

$$\sum_j w_j I(\rho_j) - I(\rho) = \int \rho(q) \operatorname{Var}(\sigma_J(q) \mid Q = q) dq \geq 0. \tag{119}$$

If each branch has zero momentum mean and assigned variance $\kappa^2 I(\rho_j)$, passive removal of the label retains their mixture variance. Resetting it to $\kappa^2 I(\rho)$ erases κ^2 times (119). A physical score-based law must carry this excess when records become unread. Equations (118)–(119) are written score calculations, not a derived recording dynamics or quantum state space. Positive distributions can reproduce these selected covariance data; the full-state obstruction of the finite quantum correction remains distinct.

Local phase data and unread energy

The score budget and unread-label excess in the score and unread-energy identities are necessary data for that candidate rule. They do not specify the evolving body state. The following finite construction tests this claim in the **supplied quantum reference** with $\hbar = 2\kappa > 0$, rather than deriving that reference or its positive scale. Its mechanism is separation and reunion of two packets, with a quantitative terminal-record bound.

Choose a nonnegative even $\phi \in C_c^\infty(-d, d)$, $\|\phi\|_2 = 1$, where $d > 0$ is a length, and fix a length $a > d$. For example normalize $e^{-1/(1-(q/d)^2)} 1_{|q|<d}$. Put

$$\phi_R(q) = \phi(q - a), \qquad \phi_L(q) = \phi(q + a), \qquad \psi_\pm = (\phi_R \pm \phi_L)/\sqrt{2}. \tag{120}$$

Their supports are disjoint. Both preparations have the same density $\rho = (\phi_R^2 + \phi_L^2)/2$. Use the finite-Fisher definition

$$I(\rho) = 4 \int |\partial_q \sqrt{\rho}|^2 dq = 4\|\phi'\|_2^2. \tag{121}$$

The score is required only ρ-almost everywhere. This definition does not divide by the zero density between the packets or at their smooth boundaries.

Proposition 17 (local-state obstruction). The two preparations in (120) have identical density-kernel jets at every diagonal point and identical expectations of every finite-order local differential observable for which they are defined. In particular all symmetrized polynomial Q, P moments agree. Nevertheless, for the supplied free Hamiltonian $P^2/(2m)$ and

$$T \geq m(a + 2d)^2/\hbar, \qquad B = \int \phi(q) dq > 0,$$

their terminal coordinate laws satisfy

$$\|\rho_{+,T} - \rho_{-,T}\|_{\mathrm{TV}} \geq \frac{2dmB^2\cos(1/2)}{\pi\hbar T} > 0. \tag{122}$$

Here total variation is the supremum of event-probability differences. Consequently an evolution rule on the equivalence classes defined by these local data cannot reproduce this reference's coordinate records.

Proof. The difference of the density kernels is $\phi_R(q)\phi_L(q') + \phi_L(q)\phi_R(q')$. It is supported in two rectangles a positive distance from the diagonal. Every diagonal derivative therefore vanishes. Also a finite-order local differential operator preserves the support of each smooth packet, so its cross-packet matrix elements vanish. This proves all the stated expectation equalities, including finite polynomial moments with $P = -i\hbar\partial_q$. These equalities do not assert equality of the full momentum distributions; finite local moments need not determine those distributions in this family.

For $T > 0$ the free propagator is

$$U_T(x,q) = \sqrt{\frac{m}{2\pi i\hbar T}}\, e^{im(x-q)^2/(2\hbar T)}.$$

Write $R_T = U_T\phi_R$, $L_T = U_T\phi_L$. Direct expansion gives $\rho_{+,T} - \rho_{-,T} = 2\,\mathrm{Re}(R_T\overline{L_T})$. For $|x| \leq d$ and either packet's integration variable, $|x - q| \leq a + 2d$. The stated time bound puts each kernel phase in $[0, 1/2]$, so the cross-phase difference is in $[-1/2, 1/2]$. Since ϕ is nonnegative, the double integral gives

$$\rho_{+,T}(x) - \rho_{-,T}(x) \geq \frac{mB^2\cos(1/2)}{\pi\hbar T} \quad (|x| \leq d).$$

Integration over this coordinate-record event proves (122), without an extra factor $1/2$. Parity also gives $\rho_{-,T}(0) = 0$. The bound makes $\rho_{+,T}(0) > 0$. □

The spatial mixture has no Fisher excess:

$$I\left(\frac{\phi_R^2 + \phi_L^2}{2}\right) = \frac{I(\phi_R^2) + I(\phi_L^2)}{2}.$$

This is a **decomposition of the position density**, not an unread which-packet instrument applied to the coherent preparations. Such an instrument would remove their cross-packet kernel entries. The zero mixture excess shows that position-score bookkeeping alone does not detect those entries; it does not identify coherent preparations with an incoherent mixture.

The missing data in a Fisher kinetic description. Consider the conditional state functional

$$H[\rho, S] = \int \left[\frac{\rho(S')^2}{2m} + \frac{\kappa^2}{2m}\frac{(\rho')^2}{\rho} + V\rho\right] dq,$$

with the Fisher term defined by (121) at zero-density boundaries. Encoding $\psi = \sqrt{\rho}\, e^{iS/\hbar}$ distinguishes (120) by $S = 0$ on both packets or by a relative constant $\pi\hbar$. Their S' and all local Hamiltonian densities agree initially. A smooth change between the two constants can lie wholly in the vacuum interval, with zero density-weighted kinetic cost. Local gradient data cannot recover this integration constant. Supplying (ρ, S) with its relative constants retains it; quotienting out independent constants on separated supports loses it. Only the common global phase can be discarded throughout this reference family. Neither this functional nor its local equations provide a continuation through the vacuum that independently selects the missing constant.

An exact two-packet completion. Retain the populations $1/2, 1/2$ and one coherence datum

$$C = \begin{pmatrix} 1/2 & c \\ \bar{c} & 1/2 \end{pmatrix}, \qquad |c| \leq 1/2. \tag{123}$$

The eigenvalues $1/2 \pm |c|$ make this a positive trace-one matrix. In the freely transported orthonormal basis (R_T, L_T), keep C fixed. Then the complete family of coordinate laws is

$$\rho_{C,T}(x) = \tfrac{1}{2}(|R_T(x)|^2 + |L_T(x)|^2) + 2\,\mathrm{Re}[cR_T(x)\overline{L_T(x)}]. \tag{124}$$

Positivity and normalization follow from the matrix and orthonormality. The free semigroup composes this description under every time subdivision. Initially every c has the same diagonal jets and Fisher data, because the packet supports are disjoint. The values $c = \pm 1/2$ give (120); $c = 0$ gives the unread which-packet mixture. This supplies the missing descriptor for this finite reference construction, not a physical derivation of complex coherence or a universal recording law.

The vacuum interval in the local-phase counterexample is not essential to loss of uniform stability. Retain its supplied $\hbar = 2\kappa > 0$ free reference. Put $u = \psi_+$ and choose a normalized smooth strictly positive Gaussian g. Let χ be smooth, equal to one on the left packet and zero on the right, with $\operatorname{supp}\chi'$ compactly inside their gap. Set $M_\chi = \|\chi'\|_\infty$ and

$$\begin{aligned} n_\epsilon &= \|u + \epsilon g\|_2, \qquad & r_\epsilon &= (u + \epsilon g)/n_\epsilon, \\ \psi_{+,\epsilon} &= r_\epsilon, & \psi_{-,\epsilon} &= r_\epsilon e^{i\pi\chi}, \qquad \epsilon > 0. \end{aligned} \tag{125}$$

Here $1 \le n_\epsilon \le 1 + \epsilon$, since u, g are nonnegative. The two densities are identical and strictly positive. Their position scores and Fisher information are exactly identical and finite.

The currents and kinetic energies instead satisfy the exact identities

$$\begin{aligned} j_{-,\epsilon} - j_{+,\epsilon} &= \frac{\pi\hbar\epsilon^2}{m n_\epsilon^2} g^2 \chi', \\ \|j_{-,\epsilon} - j_{+,\epsilon}\|_1 &\le \frac{\pi\hbar M_\chi}{m}\epsilon^2, \\ E_{\mathrm{kin},-,\epsilon} - E_{\mathrm{kin},+,\epsilon} &= \frac{\pi^2\hbar^2\epsilon^2}{2m n_\epsilon^2}\int g^2(\chi')^2 dq \le \frac{\pi^2\hbar^2 M_\chi^2}{2m}\epsilon^2. \end{aligned} \tag{126}$$

Indeed phase derivatives occur only where u and all its derivatives vanish. Every fixed diagonal kernel jet also differs by $O(\epsilon^2)$. Explicitly, write $J_{mn}(\psi) = \psi^{(m)}\overline{\psi}^{(n)}$ and let B_χ contain the compact transition region. Then

$$\|J_{mn}(\psi_{-,\epsilon}) - J_{mn}(\psi_{+,\epsilon})\|_1 \le \epsilon^2 C_{mn},$$

where the finite constant is

$$C_{mn} = \left\| 1_{B_\chi}\left[\partial^m(ge^{i\pi\chi})\partial^n(ge^{-i\pi\chi}) - g^{(m)}g^{(n)}\right]\right\|_1.$$

Outside this region the phase is constant and cancels in each jet. The same argument bounds every fixed finite-order local differential expectation with smooth coefficients on that compact region.

Proposition 18 (failure of uniform local-data stability). On these states use the following bounded metric on their local data, with the fixed length d from the local-phase counterexample supplying units:

$$d_{\mathrm{loc}}(D, \widetilde{D}) = \min\{1, d^2|I - \widetilde{I}|\} + \sum_{m,n\ge 0} 2^{-m-n-2}\min\{1, d^{m+n}\|J_{mn} - \widetilde{J}_{mn}\|_1\}. \tag{127}$$

Density is J_{00} and current is determined by J_{10}. The paired input distance tends to zero as $\epsilon \downarrow 0$. For fixed T satisfying the local-phase counterexample's time bound, put $\Delta_T = 2dmB^2\cos(1/2)/(\pi\hbar T) > 0$, with $B = \int \phi$. Their terminal coordinate laws obey

$$\|\rho_{+,\epsilon,T} - \rho_{-,\epsilon,T}\|_{\mathrm{TV}} \ge \Delta_T - 4\epsilon \ge \Delta_T/2 \quad (\epsilon \le \Delta_T/8). \tag{128}$$

No uniformly continuous decoder from the metric (127) to terminal coordinate laws can reproduce the supplied free evolution on this positive-density family.

Proof. Each fixed jet difference tends to zero and the summable weights in (127) dominate all terms; the Fisher difference is zero. Also $\|r_\epsilon - u\|_2 \le 2\epsilon$ and $e^{i\pi\chi}u = \psi_-$, so $\|\psi_{\pm,\epsilon} - \psi_\pm\|_2 \le 2\epsilon$. Free unitarity preserves this bound. For normalized amplitudes v, w, Cauchy–Schwarz gives

$$\mathrm{TV}(|v|^2, |w|^2) \le \tfrac{1}{2}\|v - w\|_2\|v + w\|_2 \le \|v - w\|_2.$$

Triangle inequalities with (122) prove (128). The paired distances then contradict uniform continuity. □

Exact reconstruction is still possible at every positive tail. On a connected strictly positive density, the phase difference is

$$\Theta(q_L, q_R) = \frac{m}{\hbar}\int_{q_L}^{q_R} \frac{j}{\rho}\, dq = \theta(q_R) - \theta(q_L) \pmod{2\pi}. \tag{129}$$

For (125), across the gap this remains $-\pi$ in the minus state, even though (126)'s unweighted current norm vanishes. Thus the result is a failure of uniform stability in the specified metric, not exact nonclosure at $\epsilon > 0$ or discontinuity of amplitude evolution. A topology controlling j/ρ directly would retain the missing information and would distinguish this pair.

Metric (127) measures a proposed local retained descriptor. It does not equal complete-record operational distinguishability over all apparatus: a full momentum readout can already distinguish the separated packet phases. The result therefore tests the descriptor and its uniformity, not a law of continuity stated for complete physical records.

Bridge-phase memory has an exact local insertion law:

$$\Theta(a, c) = \Theta(a, b) + \Theta(b, c) \pmod{2\pi}.$$

Any partition of the bridge gives the same total. Retaining that total as preparation memory through the vanishing-tail limit distinguishes the two packets after their density and all fixed local jets coalesce. In the supplied pure two-packet reference it specifies $c = \frac{1}{2}e^{i\Theta(q_L,q_R)}$ in (123). This is a concrete refinement-compatible augmentation, not a derived physical phase law. The factor $m/\hbar$ converts the bridge's velocity–length integral to a dimensionless phase; its scale has not been selected independently.

The phase-gradient term in the local-phase counterexample's conditional Hamiltonian supplies a sharper test than the particular interpolation in (125). Fix a spatial interval $B = [q_L, q_R]$, mass $m > 0$ and the supplied reference $\hbar > 0$. Let ρ be continuous and strictly positive on B, with position-density units. Hold this density and $\theta(q_L)$ fixed. Write $S = \hbar\theta$ and prescribe a real lifted phase change $\eta = \theta(q_R) - \theta(q_L)$; retaining this lift is part of the boundary data. The phase-gradient energy on the bridge is

$$E_B[\theta] = \frac{\hbar^2}{2m}\int_B \rho(\theta')^2 dq, \qquad R_B = \int_B \frac{dq}{\rho(q)}. \tag{130}$$

Here R_B has length-squared units. It is a coefficient of a spatial variational problem, not the elapsed time or the Galileo cell action.

Proposition 19 (fixed-density bridge reduction). Among absolutely continuous phases with square-integrable derivative and the prescribed endpoint lift,

$$\begin{aligned} \inf E_B &= \frac{\hbar^2\eta^2}{2mR_B}, \\ \theta_*(q) &= \theta(q_L) + \frac{\eta}{R_B}\int_{q_L}^{q} \frac{ds}{\rho(s)}, \\ j_*(q) &= \frac{\hbar}{m}\rho(q)\theta_*'(q) = \frac{\hbar\eta}{mR_B}. \end{aligned} \tag{131}$$

The minimum is attained and the phase is unique with these endpoints. If only the endpoint phases modulo 2π are prescribed, minimize $(\eta + 2\pi k)^2$ over $k \in \mathbb{Z}$; opposite half-turns can then give distinct minimizing lifts.

Proof. Weighted Cauchy–Schwarz gives $\eta^2 \le (\int_B \rho\theta'^2)(\int_B \rho^{-1})$. Equality requires θ' proportional to ρ^{-1}, yielding (131). Strict positivity and compactness make this derivative bounded and admissible; strict convexity with fixed endpoints proves uniqueness. The current formula uses the reference definition from (126), rather than a newly justified physical momentum readout. □

For a finite partition of B into intervals B_i, put $R_i = \int_{B_i} \rho^{-1} dq$. Eliminating the interior phases is exactly

$$\inf_{\sum_i \eta_i = \eta} \sum_i \frac{\hbar^2 \eta_i^2}{2mR_i} = \frac{\hbar^2 \eta^2}{2m \sum_i R_i}, \qquad \eta_i = \eta \frac{R_i}{\sum_j R_j}. \tag{132}$$

The same weighted inequality proves this formula, and $R_B = \sum_i R_i$ makes its repeated elimination associative under every cut. No limiting curve or shape-regular partition is assumed in this fixed-density reduction. Its connection with the cut action is the Dirichlet minimization mechanism; the spatial density weight and energy units differ from Newton's temporal functional K_τ.

Positive-tail limit with its sharp coefficient. Choose B strictly inside the vacuum interval of (120), between the packets, and use $\rho_\epsilon = r_\epsilon^2$ from (125). On this bridge $u = 0$, so

$$R_{B,\epsilon} = \frac{n_\epsilon^2}{\epsilon^2} R_{B,g}, \qquad R_{B,g} = \int_B g^{-2} dq, \qquad \inf E_{B,\epsilon} = \frac{\hbar^2 \eta^2 \epsilon^2}{2m n_\epsilon^2 R_{B,g}}. \tag{133}$$

The fixed Gaussian g is strictly positive on the compact bridge, so $R_{B,g}$ is finite. At $\eta = -\pi$, take phase π to the left of B and zero to its right. This preserves the two opposite packet phases. The minimizing derivative on B is independent of ϵ. Extending it by zero outside gives an H^1 amplitude; if globally smooth phases flat at the bridge endpoints are required, the same value is the infimum. Indeed replace g^{-2} in the derivative by $\chi_\delta g^{-2}$, where $0 \le \chi_\delta \le 1$ is smooth, vanishes near the endpoints and tends to one in the interior, and normalize its integral to η. Dominated convergence gives (133) as $\delta \downarrow 0$. At fixed δ, all phase derivatives are fixed and finite, as required for the positive-tail calculation's local-jet comparison.

For every such interpolation, not just the original χ in (125), its multiplication by u still gives ψ_-. Consequently the same 2ϵ amplitude bound and (128)'s TV $\ge \Delta_T - 4\epsilon$ hold in the supplied free reference. Thus making the bridge optimally cheap preserves the predictive phase witness. A history cannot infer the missing datum from its bridge energy alone. The constant current in (131) is a static minimizing profile; outside the connector it supplies neither a continuity equation for the whole body nor apparatus dynamics.

The exact reconstruction estimate also identifies its conditioning. At fixed ρ an error δj in the current gives

$$|\delta\eta| \le \frac{m}{\hbar} \sqrt{R_B} \left(\int_B \frac{|\delta j|^2}{\rho} dq \right)^{1/2}. \tag{134}$$

Equality occurs for a constant current error. For (133) this sharp factor grows as $n_\epsilon \sqrt{R_{B,g}}/\epsilon$. The bound defines a weighted data norm, not an operational instrument cost. If an additional premise supplied $\rho \ge \rho_* > 0$ on a bridge of length ℓ, (131) would instead give $E_B \ge \hbar^2 \rho_* \eta^2/(2m\ell)$. The positive-density family alone supplies no uniform ρ_*. Even such a conditional energy bound would not select a universal action unit or its positivity.

What an unread label must retain when currents are present. Extend (119) to a finite ensemble of smooth positive densities and real smooth phases (ρ_j, θ_j) with weights w_j and finite kinetic energies. Put $\rho = \sum_j w_j \rho_j$, $a_j = \theta_j'$, and $\bar{a}(q) = \sum_j w_j \rho_j(q) a_j(q)/\rho(q)$. On the line, where $\bar{a}$ is locally integrable, integrate it to obtain a phase for the descriptor with this density

and averaged current. With $\sigma_j = -\partial_q \log \rho_j$,

$$\sum_j w_j E_{\mathrm{kin}}[\rho_j, \theta_j] - E_{\mathrm{kin}}[\rho, \bar{\theta}] = \frac{\hbar^2}{8m} \int \rho \, \mathrm{Var}(\sigma_J \mid Q = q) dq \\ + \frac{\hbar^2}{2m} \int \rho \, \mathrm{Var}(a_J \mid Q = q) dq \geq 0, \qquad \bar{\theta}' = \bar{a}. \tag{135}$$

Here $E_{\mathrm{kin}}[\rho, \theta] = \hbar^2 I(\rho)/(8m) + \hbar^2 \int \rho \theta'^2/(2m)$. The first term in (135) is exactly (119), with $\hbar = 2\kappa$; the second is the conditional-variance identity for phase gradients. Passive forgetting retains $\sum_j w_j E_{\mathrm{kin}}[\rho_j, \theta_j]$. Replacing the ensemble by its density/current descriptor resets both displayed excesses. Equation (135) supplies their amounts, without identifying that descriptor as the correct physical state after forgetting.

These excesses still omit relative constants between separated packets: in (120) all local gradients and densities coincide, while averaging the two preparations erases the off-diagonal coherence in (123). For fixed smooth positive-tail interpolations their local excesses are only $O(\epsilon^2)$ on the bridge. The conditional energy accounting and the retained phase history therefore impose different obligations.

Consequence for STATE

Affine copying and its continuum feedback limit are exact constructions under supplied joint preparation and declared coordinate-readout rules. Nonlinear copies and passive side access expose the limits of those rules; local score data also omit predictive phase information and unread energy. Every positive scale here is assumed. Existence of a classical body–record law, preservation of a conditional covariance floor, quantum state/instrument structure and universal calibration remain distinct obligations. Complete conjugate access, physical controller memory, energy, cutoff and aperture restrictions must be specified before any necessity claim is reused.